# F1SIP

Formula 1 Strategy Intelligence Platform

## MASTER IMPLEMENTATION SPECIFICATION

Version 3.0 · Refined, Corrected and Rebuilt from Specification v2.0

| Field | Value |
|---|---|
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| Document status | v3.0 — for your approval before any code is written |
| Prepared | 7 July 2026 |
| Replaces | F1SIP Production Specification v2.0 |
| Data access | FastF1 (Jolpica backend) + OpenF1 historical API |
| Grid model | 22 cars (11 teams incl. Cadillac) — 2026 season reality |
| Non-negotiable | Nothing is a black box: every component is inspectable and explainable |
| Execution target | Claude Code, module by module, tests first |

*Naming: your prompt calls the project SENNA; the attached v2 document calls it F1SIP. This document uses F1SIP throughout to match the submitted proposal. Confirm the final name in Open Question OQ-1.*

**What this document is.** A forensic audit of Specification v2, followed by a complete rebuilt specification you can hand to Claude Code and execute. Everything that was factually wrong has been corrected against primary 2026 sources. Everything that would have blown up later — reward hacking, dead APIs, impossible compute, trademark exposure — has been removed or replaced, with the reasoning written down so you can defend every decision in your viva.

# Contents

**Contents** . . . . . 2

**Part 0 — How to read this document (start here)** . . . . . 4

0.1 The one-paragraph story . . . . . 4

0.2 The five most important things that changed since v2 . . . . . 4

0.3 Map of this document . . . . . 4

0.4 The interpretability contract (your non-negotiable, made concrete) . . . . . 5

**Part 1 — Forensic audit of Specification v2** . . . . . 6

1.A Physics and 2026 regulation errors . . . . . 6

1.B Data pipeline flaws . . . . . 7

1.C Environment and RL design flaws . . . . . 8

1.D Backend, frontend, deployment and academic-risk flaws . . . . . 11

**Part 2 — Edge cases and failure modes (with causal mechanisms)** . . . . . 13

2.1 Simulation and regulation edge cases . . . . . 13

2.2 Data pipeline edge cases . . . . . 14

2.3 Serving and frontend edge cases . . . . . 14

**Part 3 — Removed from v2 (and exactly why)** . . . . . 15

**Part 4 — Added in v3 (and what each addition buys)** . . . . . 15

4.1 What v2 got right — carried forward unchanged . . . . . 16

**Part 5 — The rebuilt master specification** . . . . . 17

5.1 System architecture . . . . . 17

5.2 Corrected 2026 physics model (the constants that drive everything) . . . . . 17

5.3 Data pipeline — exact sources, columns and cleaning contract . . . . . 18

5.4 Synthetic 2026 generation — eight named transformations . . . . . 19

5.5 Gymnasium environment contract (F1RaceEnv) . . . . . 21

5.6 Training pipeline — MaskablePPO . . . . . 23

5.7 Curriculum — three stages with competence gates . . . . . 24

5.8 Explainability layer — what is explained, how, and how to say it . . . . . 24

5.9 Technology stack (pinned) and FastAPI backend . . . . . 26

5.10 Frontend — six views, each with its assumptions stated . . . . . 27

5.11 Deployment — local compose for dev, Railway services for prod . . . . . 28

5.12 Validation protocol — what each metric measures, and how it lies . . . . . 29

5.13 Recruiter demo — the first 90 seconds, engineered . . . . . 29

**Part 6 — Beginner build guide (plain language, step by step)** . . . . . 31

6.1 How to drive Claude Code with this document . . . . . 31

6.2 The nine phases . . . . . 31

6.3 The Minimum Defensible Product and the cut order . . . . . 32

**Part 7 — Open questions (your answers unblock the build) . . . . . 33**

**Part 8 — Assumptions register (2026 physics and data) . . . . . 33**

Approval checklist — what happens next . . . . . 34

# Part 0 — How to read this document (start here)

This part is written in plain language, because you asked to be guided step by step. Everything after Part 0 is written as a formal engineering specification, because that is what your examiner, your supervisor and a motorsport hiring manager will judge.

## 0.1 The one-paragraph story

You are building a race strategist made of software. A **simulator** plays out a full Formula 1 race, lap by lap, with 22 cars, tyres that wear out, a battery that charges and drains, weather, and safety cars. An **agent** (a neural network trained with reinforcement learning) sits in one of those cars and decides, every lap: stay out or pit, which tyre, and how to spend battery energy. After training on hundreds of thousands of simulated races, the agent gets good. You then **prove** it is good with careful experiments, **explain** every decision it makes with SHAP-based decision cards, and **show** it all in a 3D dashboard anyone can open in a browser.

## 0.2 The five most important things that changed since v2

- **1. The physics in v2 was wrong, and it is now fixed.** v2 modelled a "400 kW MGU-K" and a "4 MJ per lap deployment cap". Neither exists. The real 2026 rules: the electric motor (MGU-K) delivers **350 kW** (400 kW is the petrol engine), the battery stores **4 MJ** (not 8), and the binding limits are a **harvest cap of about 8.5 MJ per lap** (circuit-dependent, 5-9 MJ) plus a **speed taper** on deployment. Building the dissertation on the wrong numbers would have been fatal in the viva. Section 5.2 has the corrected model with sources.
- **2. Rule-breaking is now impossible instead of merely punished.** v2 let the agent break energy rules and fined it with a -8 reward, then celebrated "99.5% compliance". v3 uses action masking and physical clipping so illegal actions cannot happen at all: **100% compliance by construction** — a much stronger engineering statement, and it frees the reward function to be simple and honest.
- **3. The reward function shrank from 12 terms to 3.** Several v2 terms actively coached the agent toward the conclusions you wanted to "discover" (a bonus for pitting under safety car, a bonus for "ERS efficiency"). That is circular: you cannot claim the agent learned something you paid it to do. Those behaviours are now **measured**, not rewarded.
- **4. Dead and paid data sources are gone.** Ergast shut down at the start of 2025 — v2 still listed it. FastF1 3.7+ now reads the Jolpica replacement automatically. OpenF1 historical data (2023 onward) is free; its real-time stream is paid, so the "live session" feature now runs on your own simulator instead.
- **5. The 2026 season already exists — use it.** Roughly ten real 2026 rounds have been raced as of today. v2 planned to invent 2026 data from physics guesses. v3 **calibrates and validates** the 2026 model against real 2026 races. This turns your weakest chapter into your strongest one.

## 0.3 Map of this document

| Part | What it contains | Read it when |
|---|---|---|
| Part 1 | Forensic audit: 30 numbered findings (F-01 to F-30), each with severity, cause and fix | Before approving — this is the evidence for every change |
| Part 2 | Edge cases and failure modes with causal mechanisms (E-01 to E-18) | When building the simulator and the frontend |
| Part 3 | What was removed and why (R-01 to R-12) | Before approving |
| Part 4 | What was added and why (A-01 to A-12), plus what v2 got right and is kept | Before approving |
| Part 5 | The rebuilt specification: physics, data, environment, training, explainability, backend, frontend, deployment, validation, demo | During the build — hand this to Claude Code |
| Part 6 | Beginner build guide: 9 phases in plain language, re-baselined timeline from 8 July, and a cut-order if you fall behind | Every Monday morning |
| Part 7 | Open questions OQ-1 to OQ-12 — your answers unblock the build | Right now |

| Part | What it contains | Read it when |
|---|---|---|
| Part 8 | Assumptions register: every 2026 physics number, its source and confidence | When writing Chapter 6 and preparing the viva |

## 0.4 The interpretability contract (your non-negotiable, made concrete)

"Nothing is a black box" only means something if each component has a named inspection mechanism. This table is that contract. If a component cannot satisfy its row, it must be redesigned or replaced.

| Component | How it is inspectable | Artifact produced |
|---|---|---|
| Tyre / fuel / energy physics | Closed-form equations with named coefficients fitted from data; every fit has a residual plot and a config file entry | physics_params.yaml + calibration notebook per module |
| Reward function | Three terms only; per-lap reward decomposition logged for every episode | reward_trace.parquet per evaluation race |
| Action legality | Explicit mask function with unit tests; mask reasons logged per step | mask_reasons column in decision log |
| Policy decisions | SHAP attribution on a distilled surrogate with reported fidelity, cross-checked by gradient attribution on the true network | Decision cards (top-5 features + counterfactual flip point) |
| Curriculum transitions | Numeric gates evaluated on frozen seed sets; gate values logged | curriculum_gates.json + W&B panel |
| Rival behaviour | Three archetypes as readable rule sets with published parameters and a train/test parameter split | rivals.yaml |
| Synthetic 2026 data | Eight named transformations, each with formula, source and a validation plot against real 2026 rounds | Chapter 6 figures + transforms.yaml |
| Evaluation metrics | Formal definition, denominator, and a written "how this metric can be gamed" note for every metric | Section 5.12 of this document |
| Frontend claims | Every number shown in 3D traces back to an API response traceable to a database row | API response IDs rendered in dev overlay |

*This table doubles as a viva answer. When asked "is any of this a black box?", you walk the examiner down this contract row by row.*

# Part 1 — Forensic audit of Specification v2

Thirty findings, ordered by category then severity. Each states exactly what v2 says, the causal reason it is wrong or dangerous, and the replacement adopted in Part 5. Severity: CRITICAL = would invalidate results or fail the viva; HIGH = would cost weeks or produce misleading claims; MEDIUM = quality/robustness; LOW = polish.

## 1.A Physics and 2026 regulation errors

### F-01 [CRITICAL] The MGU-K is 350 kW, not 400 kW

| | |
|---|---|
| **v2 says** | Pillar 1, Section 4.5 and Claim 1 all state a "400 kW MGU-K" scaled as "120 kW x 3.33". |
| **Why it is a problem** | The 2026 FIA regulations set MGU-K electrical output at **350 kW** (about 470 hp); **400 kW is the ICE** (petrol engine) figure, reduced from roughly 550-560 kW. v2 has confused the two halves of the power unit. Every downstream number — deployment budgets, straight-line speed scaling, the "x3.33" factor — inherits the error, and the headline novelty claim ("first system implementing the 400 kW MGU-K") would be dismantled by any examiner or F1 engineer in one sentence. |
| **Fix (v3)** | All energy math rebuilt around MGU-K = 350 kW with the official speed taper (full power to 290 km/h, linear taper to zero at 355 km/h) and Manual Override (350 kW to 337 km/h, roughly +0.5 MJ, available within 1.0 s of the car ahead). Sources: Formula1.com 2026 PU explainers, FIA 2026 PU Technical Regulations Art. 5, ESPN 2026 rules guide. See Section 5.2 and the Assumptions Register. |

### F-02 [CRITICAL] The "4 MJ/lap hard deployment cap" does not exist

| | |
|---|---|
| **v2 says** | The 4 MJ/lap cap is called "the defining technical constraint of the project" (Section 13.2); the ERS view draws a red cap plane at 4 MJ; the reward has a -8.0 violation penalty; a target metric is "99.5% cap compliance". |
| **Why it is a problem** | 4 MJ is the **usable capacity of the Energy Store (battery)** — unchanged for 2026 — not a per-lap deployment limit. Because the car also harvests during the lap (braking, lift-and-coast at 350 kW, "super clipping" on throttle at 250 kW), per-lap deployment routinely exceeds 4 MJ; the true binding constraints are (i) battery state of charge, (ii) the 350 kW power ceiling with its speed taper, and (iii) a **harvest** cap of 8.5 MJ/lap baseline, adjusted per circuit by the FIA (range about 5-9 MJ; e.g. reduced at Monza, raised at Monaco). A whole view, a reward term, a unit test and a headline metric are built on a rule that is not in the rulebook. |
| **Fix (v3)** | Energy model rebuilt in Section 5.2: SOC in [0, 4] MJ; harvest cap H_max(circuit) as a config parameter defaulting to 8.5 MJ; deployment limited by a per-circuit deployable-energy integral derived from the power taper. The ERS Landscape view now shows SOC trajectory plus harvest/deploy bars against the per-circuit harvest cap — which is both correct and more interesting. |

### F-03 [CRITICAL] Battery capacity is 4 MJ, not 8 MJ

| | |
|---|---|
| **v2 says** | Section 4.5: "Battery capacity ~4 MJ becomes ~8 MJ estimated (team technical announcements)". State feature #6 has range 0-8 MJ. |
| **Why it is a problem** | The usable energy store remains capped at 4 MJ in 2026; with a 350 kW motor this means a full battery can be emptied roughly three times faster than before — which is precisely why energy management is the strategic story of 2026. Doubling the battery in the simulator removes the scarcity that creates the whole decision problem: the agent would learn strategies that are systematically too aggressive to transfer to reality. |
| **Fix (v3)** | SOC range corrected to [0, 4] MJ everywhere (observation, physics, UI axes). The scarcity dynamics — deplete-then-derate — are modelled explicitly (Section 5.5, E-01). |

### F-04 [CRITICAL] Harvest model misses half of the 2026 energy game

| | |
|---|---|
| **v2 says** | Section 5.1 models harvesting only "per braking zone" from a circuit energy map. |
| **Why it is a problem** | 2026 cars recover energy three ways: regenerative braking, **lift-and-coast** (350 kW, but it forces the active aero out of low-drag Straight mode — a real lap-time cost), and **super clipping** (automatic recharge at up to 250 kW while at full throttle). The strategic trade-off drivers actually face in 2026 — sacrifice time now to bank energy for later — lives in these mechanisms. A braking-only model makes the "conserve" action nearly free, so the agent would learn an energy strategy with no real-world counterpart, and the "ERS debt" novelty claim would describe an artefact of the simulator. |
| **Fix (v3)** | Harvest is modelled as a per-mode quantity with an explicit time cost: conserve mode adds lift-and-coast harvest and pays a pace penalty (drag + earlier lift); balanced mode uses braking + clipping only. Formulas and calibration in Section 5.5, transformation T4 in 5.4. |

### F-05 [HIGH] Active aero is not the overtaking aid; Manual Override is

| | |
|---|---|
| **v2 says** | Pillar 1 and Section 5.1: "active aero replacing DRS", with "ERS state gates drag reduction on defined straight sections" and state feature #20 "active aero available = overtake feasibility". |
| **Why it is a problem** | X-mode (low drag) / Z-mode (high downforce) active aero is available to **every** car on designated sections regardless of gaps — it is not a following-car advantage and it is not gated by ERS state. The overtaking aid that replaced DRS is **Manual Override** (the Boost button): within 1.0 s at a detection point, the chasing car keeps 350 kW up to 337 km/h while the leader tapers from 290 km/h, worth about 0.5 MJ. v2 therefore couples two systems the rulebook keeps separate, and the frontend would visualise a mechanism that does not exist. |
| **Fix (v3)** | Aero model: X/Z mode as a per-section property affecting drag for all cars, disabled during lift-and-coast. Overtake feasibility feature replaced by override_available (gap under 1.0 s). Circuit Explorer legend updated. See Section 5.2 and 5.10. |

### F-06 [HIGH] Fuel numbers belong to the old rules

| | |
|---|---|
| **v2 says** | Section 5.1: fuel starts "~100 kg (2026), burns ~1.75 kg/lap". |
| **Why it is a problem** | 2026 cars are energy-flow limited on sustainable fuel with a race allowance far below 100 kg (order of 70 kg); carrying 30 kg of phantom fuel distorts the weight model by roughly a second a lap early in the race (at ~0.03 s/kg) and mis-prices the fuel-save trade-off. Also note minimum car mass is 768 kg (-30 kg), not "-20 kg". |
| **Fix (v3)** | Fuel initialised from a per-circuit energy budget with mass ~70 kg flagged VERIFY in the Assumptions Register (check FIA 2026 Technical Regulations before Phase 2); mass corrected to 768 kg; weight sensitivity fitted from data rather than fixed at 0.035 s/kg (Section 5.4, T6). |

### F-07 [HIGH] The fastest-lap point was abolished — action 11 rewards a ghost rule

| | |
|---|---|
| **v2 says** | Action 11 "Push for fastest lap (laps 65-70, P1-P10)" plus reward component "+1.0 if fastest lap claimed", described as a "bonus point replica". |
| **Why it is a problem** | Formula 1 removed the fastest-lap championship point from the 2025 season onward. The agent would spend late-race energy chasing a point that no longer exists, and the examiner question "why does your 2026 agent optimise a 2019-2024 rule?" has no good answer. |
| **Fix (v3)** | Action and reward term deleted (R-01). Late-race pace remains available through the ordinary attack mode. Flagged VERIFY in the Assumptions Register (30-second check against 2026 Sporting Regulations). |

### F-08 [HIGH] Grid, calendar and circuit count are out of date

| | |
|---|---|
| **v2 says** | Race model "20-22 cars"; "all 24 circuits"; Stage 3 trains on "all 24 circuits"; instancing notes say "19 rival geometries". |
| **Why it is a problem** | The 2026 grid is **22 cars** (Cadillac joined as the 11th team). The 2026 calendar is currently **22 rounds** after two cancellations, includes the brand-new Madrid street circuit (September — no telemetry exists for it until then) and drops Imola. Hard-coding "24" anywhere produces off-by-N bugs in position vectors, instancing counts and coverage claims. |
| **Fix (v3)** | Grid fixed at 22 (21 rivals); circuit library is data-driven: circuits.yaml lists every circuit for which validated GPS + lap data exists, and every claim reads "all N circuits with validated data" where N is computed, not asserted. Madrid enters the library automatically once its first session is ingested. |

### F-09 [MEDIUM] "ERS debt" is the wrong mental model; SOC scarcity is the right one

| | |
|---|---|
| **v2 says** | Pillar 2 and Claim 4 describe "ERS debt patterns where deliberate underdeployment early unlocks a decisive end-race attack window", framed around the per-lap cap. |
| **Why it is a problem** | With no per-lap deployment cap, "debt" against that cap is undefined. The real 2026 mechanism is richer: a 4 MJ battery, a 350 kW drain, and harvest that costs lap time creates a genuine inter-temporal banking problem — but it is about **state of charge** and **where on the calendar of the lap you harvest**, not about a cap ledger. Writing the claim in cap language invites a correct and devastating "that rule does not exist" rebuttal. |
| **Fix (v3)** | Claim reworded as "emergent energy banking": the agent trades lap time for SOC early to buy attack laps later. Measured via the SOC trajectory and mode sequence; the phenomenon (if it emerges) is now defensible because the mechanism exists in the rulebook. |

## 1.B Data pipeline flaws

### F-10 [CRITICAL] Ergast is dead; v2 lists it as a live backup

| | |
|---|---|
| **v2 says** | Tech stack row: "Ergast API — Legacy — race results, standings, circuit metadata backup". |
| **Why it is a problem** | Ergast shut down at the beginning of 2025. Any code path that calls ergast.com fails today. The community successor is **Jolpica-F1** (drop-in compatible), which FastF1 uses automatically from version 3.7. |
| **Fix (v3)** | Ergast removed from the stack (R-04). Pin fastf1>=3.7. Results/standings come through FastF1's Ergast-compatible interface (served by Jolpica) with its rate limits respected (bulk ingest once, store in PostgreSQL, never query live in the serving path). |

### F-11 [HIGH] OpenF1 realities: 2023 onward only, and real-time is paid

| | |
|---|---|
| **v2 says** | Tech stack: OpenF1 for "live session data, real-time timing, car telemetry streams"; data scope table implies OpenF1 supplements all seasons. |
| **Why it is a problem** | OpenF1 provides data from **2023 onward** (nothing for 2022) and its true real-time stream (MQTT/WebSocket during sessions) now requires a paid account; historical REST access is free at about 3 req/s. Two consequences: (i) 2022 pit-duration and interval features must come from FastF1 alone; (ii) any feature advertised as "live F1 data" either costs money or silently shows stale data during a race weekend. |
| **Fix (v3)** | OpenF1 is scoped to historical enrichment (2023+ pit durations, intervals, race control). The "Live Simulation" view is explicitly the **agent racing in the simulator**, streamed over your own WebSocket — no OpenF1 dependency (R-05). Ingestion throttled to the free tier. |

### F-12 [HIGH] Circuit geometry: wrong FastF1 objects, wrong projection, noisy Z

| | |
|---|---|
| **v2 says** | Section 4.4: "fetch GPS lat/long per metre from get_circuit_info(), project with Mercator, elevation from GPS altitude channel". |
| **Why it is a problem** | fastf1.get_circuit_info() returns corner/marshal metadata, not a dense centreline. The dense track outline comes from **position telemetry** (X, Y, Z of a clean fast lap), which is already in a local metric frame — Mercator projection of lat/long is both unnecessary and unavailable. The Z channel is noisy and sometimes absent; extruding it raw produces a rollercoaster ribbon. Banking "derived from corner radius and speed" is an invented quantity with no data source and should be labelled as stylised. |
| **Fix (v3)** | Pipeline rebuilt (Section 5.10): pick fastest clean lap, take pos_data X/Y/Z, resample by arc length at 2 m, smooth with Savitzky-Golay, close the loop, build rotation-minimizing frames (prevents ribbon twist at hairpins), extrude 12 m width, exaggerate smoothed elevation x1.5 and label it stylised. Corner metadata from get_circuit_info() drapes annotations on top. |

### F-13 [MEDIUM] ERS "proxy from throttle-off" oversells what telemetry contains

| | |
|---|---|
| **v2 says** | Section 4.2: FastF1 telemetry used for "ERS proxy derivation (throttle-off = harvest)". |
| **Why it is a problem** | No public channel exposes battery SOC or MJ flows; throttle-off time is a weak correlate that also captures braking, lift for corners and traffic. Presenting it as measured ERS invites the examiner question "where does this channel come from?" with an awkward answer. |
| **Fix (v3)** | Reframed honestly: energy quantities are **simulator-internal**, with per-circuit deployable/harvestable energy integrals estimated from speed-distance telemetry (braking energy from kinetic-energy deltas), calibrated so aggregate behaviour matches 2026 broadcast facts (e.g. cars visibly derating). Chapter 6 states this limitation up front — honesty here earns marks. |

### F-14 [MEDIUM] Known FastF1 quirks are unhandled in the ingest plan

| | |
|---|---|
| **v2 says** | Week 2 says "all sessions loaded, cleaned, stored" without naming the cleaning rules. |
| **Why it is a problem** | Real FastF1 data has systematic traps: LapTime is NaT on in/out laps; IsAccurate flags unreliable laps; TrackStatus is a digit string needing parsing ("4" SC, "6" VSC, "5" red); sprint weekends change session naming; crashed cars leave truncated stints; some 2022 sessions have missing telemetry channels; position data is sampled at ~4-5 Hz with occasional dropouts. Unhandled, each one becomes a silent bias in the fitted tyre and pace models. |
| **Fix (v3)** | Section 5.3 contains an explicit cleaning contract (per-column rules and exclusion filters) and a data-quality report that must be produced per season before any model fitting. |

## 1.C Environment and RL design flaws

### F-15 [CRITICAL] The position reward can be farmed: +1.0 gain vs -0.8 loss

| | |
|---|---|
| **v2 says** | Reward: "+1.0 per position gained, -0.8 per position lost — asymmetric, preserves risk-taking incentive". |
| **Why it is a problem** | Any policy that causes its position to oscillate collects +0.2 per swap cycle: lose a place (-0.8), regain it (+1.0). Pitting into traffic and re-passing backmarkers becomes a reward pump that has nothing to do with finishing higher. This is a textbook reward-hacking hole, and it also double-counts with the terminal position bonus. |
| **Fix (v3)** | Replaced with **potential-based shaping**: F = gamma * Phi(s') - Phi(s) with Phi(s) = -0.1 x position. Ng, Harada & Russell (1999) prove this leaves the optimal policy unchanged while still giving per-lap credit signal; the sum telescopes so oscillation earns exactly zero. Section 5.5. |

### F-16 [CRITICAL] Penalising rule breaches (then reporting 99.5% compliance) is the wrong architecture

| | |
|---|---|
| **v2 says** | ERS breach handled by clipping plus a -8.0 penalty; target metric "ERS compliance > 99.5%". |
| **Why it is a problem** | If the simulator can enter illegal states, three bad things follow: (i) training wastes capacity learning "do not press the illegal button"; (ii) 0.5% of a 60-lap race is still a disqualification in reality, so the metric celebrates an agent that would be excluded from results; (iii) the metric is trivially gameable by shrinking the action space late in training. A senior engineer reads "99.5% compliant" as "0.5% illegal". |
| **Fix (v3)** | Constraints move into the environment: **action masking** (MaskablePPO from sb3-contrib) for discrete legality (pit lane closed, compound unavailable, wet rules) and **physical clipping inside the dynamics** for continuous limits (cannot deploy energy the battery does not hold; harvest saturates at H_max). Compliance is 100% by construction, verified by a 100k-random-action property test. The interesting learned quantity becomes energy **efficiency**, reported instead (Section 5.12). |

### F-17 [CRITICAL] Several reward terms pre-write the conclusions

| | |
|---|---|
| **v2 says** | Rewards: +2.5 for pitting in the first 2 laps of a safety car; +0.1 x (MJ deployed / "circuit-optimal MJ"); +0.2 x grip remaining at stint end; +0.5 for a successful undercut/overcut. |
| **Why it is a problem** | Each term injects the designer's answer into the objective, which destroys the corresponding claim: you cannot report "the agent learned to exploit safety cars 78% of the time" when you paid it 2.5 points to do so — the metric then measures reward-following, not strategy. The "circuit-optimal MJ" term is circular (it presumes the optimum the agent is meant to find). The grip-at-stint-end bonus is perverse: it pays the agent for pitting early on barely-worn tyres. The undercut bonus needs a hand-written success-labelling rule, which smuggles in more assumptions. |
| **Fix (v3)** | All four removed (R-02). Safety-car exploitation, energy efficiency, tyre management and undercut success become **evaluation metrics** with precise definitions (Section 5.12). The reward is 3 terms: small pace shaping, potential-based position shaping, terminal points. A dedicated ablation trains v2's 12-term reward as a comparison — likely a publishable negative result on reward over-engineering. |

### F-18 [HIGH] A flat 12-action space entangles two independent decisions

| | |
|---|---|
| **v2 says** | Twelve mutually exclusive actions mixing stay-out ERS modes, five pit-compound choices and a fastest-lap action. |
| **Why it is a problem** | Pit choice and energy mode are independent axes; flattening them means (i) the agent cannot express "pit AND attack the out-lap", (ii) all six stay-out actions compete as unrelated siblings so credit assignment is noisier, and (iii) legality rules become a tangle of per-action conditionals instead of two clean masks. |
| **Fix (v3)** | MultiDiscrete([6, 4]): head A pit decision {stay, pit-S, pit-M, pit-H, pit-Inter, pit-Wet}; head B energy/pace mode {conserve, balanced, attack, defend}. Per-head mask functions; MaskablePPO supports MultiDiscrete masking natively. Also the honest reason DQN leaves the ablation set: SB3 DQN requires a flat Discrete space (F-22). |

### F-19 [HIGH] Observation vector: leakage in, key facts out

| | |
|---|---|
| **v2 says** | Feature #22 "target position delta" described as "contextual reward shaping"; #7 "ERS deployed this lap" (a within-lap quantity at lap granularity); no feature for the mandatory two-compound rule, pit-stop count, laps remaining as an explicit scalar, harvest budget or rival tyre context beyond three binary "archetype pitted" flags. |
| **Why it is a problem** | Putting a reward-shaping target inside the observation leaks objective information the real strategist would not have and makes SHAP explanations circular ("it pitted because the target said so"). Meanwhile the agent cannot see the single most binding sporting rule (must use two dry compounds — without it, the learned optimum is a zero-stop race) nor the opponent tyre state that drives real undercut decisions. |
| **Fix (v3)** | Observation rebuilt as a documented 32-dimension vector (Section 5.5): adds second_compound_used flag, pit_count, laps_remaining, harvest headroom, gaps plus tyre-age/compound deltas to the cars directly ahead and behind, override availability; removes #22 and #7. Every dimension has a name, range, source and rationale row. |

### F-20 [HIGH] The mandatory two-compound rule is missing entirely

| | |
|---|---|
| **v2 says** | Neither the action constraints, the state space, nor the edge-case table mention the rule that a driver must use two different slick compounds in a dry race. |
| **Why it is a problem** | This rule is the reason pit stops exist at all in dry races. Without it the globally optimal strategy in the simulator is frequently "never pit", the agent learns it, and every counterfactual comparison against real strategies (which obey the rule) is meaningless. |
| **Fix (v3)** | Enforced structurally: if the race is dry and the agent has used one compound, the stay-out action is masked on the final feasible pit lap (and results mark non-compliance as classification-equivalent failure in evaluation). second_compound_used is an observation feature. Unit test added (Section 5.5 termination/legality table). |

### F-21 [HIGH] Curriculum advances on a step budget, not on competence

| | |
|---|---|
| **v2 says** | Stages transition after fixed 5M / 7M / 8M steps. |
| **Why it is a problem** | Fixed budgets either waste compute (agent mastered Stage 1 at 2M) or promote a failing agent (entropy collapsed at 4.8M); either way the ablation "no curriculum" is confounded because stage time was arbitrary. Examiners ask "why 5M?" and "because the plan said so" is not an answer. |
| **Fix (v3)** | Competence-gated transitions with numeric criteria evaluated on frozen seed sets (Section 5.7): Stage A exits on plateau AND pace within 1.5% of a computable solo-race reference; Stage B exits on paired-seed superiority over the scripted baseline. Budgets become ceilings, not schedules. |

### F-22 [MEDIUM] The 8-experiment ablation grid is compute-impossible and partly incoherent

| | |
|---|---|
| **v2 says** | Eight ablations, each implicitly at the full 20M-step budget, including DQN and A2C baselines, plus "low gamma" and "no SC model". |
| **Why it is a problem** | 8 x 20M = 160M additional environment steps — months of compute for one student. DQN cannot run at all after the MultiDiscrete refactor (SB3 DQN needs Discrete). "No SC model" changes the environment, so its reward numbers are not comparable to the others without careful framing. |
| **Fix (v3)** | Ablation set cut to four high-value runs at a reduced 4M budget each, 2 seeds (Section 5.6): (1) no curriculum; (2) penalty-based constraints instead of masking (tests F-16 directly); (3) v2's 12-term reward vs the minimal reward; (4) terminal-only reward (no shaping). A2C kept as an optional fifth if compute allows; DQN dropped with the documented reason — itself a good viva answer. |

### F-23 [MEDIUM] VecNormalize statistics are a silent deployment bug waiting to happen

| | |
|---|---|
| **v2 says** | VecNormalize is enabled for training; nothing says the running mean/var must be saved, frozen and shipped with the model. |
| **Why it is a problem** | If inference normalises with fresh (or no) statistics, the policy receives inputs on a different scale than it was trained on and degrades quietly — the classic "great in W&B, random in the demo" failure. Curriculum stage changes also shift observation distributions mid-run. |
| **Fix (v3)** | vecnormalize.pkl is saved with every checkpoint, loaded with training=False at inference, and its hash is part of the model version record. Stage transitions keep the running statistics (documented choice). Integration test: same state, training-eval vs served-API action must match. |

### F-24 [MEDIUM] Counterfactual replays mix eras: a 2026-physics agent vs 2023/24 races

| | |
|---|---|
| **v2 says** | The 10 counterfactual races are 2023-2024 events; the agent is trained under 2026 physics; positions gained are compared directly. |
| **Why it is a problem** | The comparison confounds strategy quality with regulation change: the agent "wins" partly because it drives a different car. An examiner will ask which effect you measured. |
| **Fix (v3)** | The environment takes an **era config** (2024 physics vs 2026 physics — same code, different constants file). Counterfactual chapter runs era-matched: a quick Stage-C fine-tune on 2024 constants replays the 2023/24 races; the headline 2026 agent is evaluated on 2026 rounds and synthetic 2026 races. Decision recorded as OQ-9. |

### F-25 [MEDIUM] SHAP plan names the wrong explainer and an infeasible budget

| | |
|---|---|
| **v2 says** | Section 6.5: "fit SHAP TreeExplainer (or KernelExplainer) on the policy network's action probability outputs" across 1000 evaluation episodes. |
| **Why it is a problem** | TreeExplainer only works on tree models — the policy is an MLP, so that call raises immediately. KernelExplainer needs thousands of model evaluations per explained decision; at 1000 episodes x ~55 decisions x 32 features that is on the order of a billion forward passes. The fallback would be silently explaining 50 laps and captioning it as 55,000. |
| **Fix (v3)** | Two-path design (Section 5.8): distill the policy into a gradient-boosted surrogate with **reported fidelity** (top-1 agreement target >= 90%) and run exact TreeSHAP everywhere cheaply; cross-check on 500 sampled decisions with GradientExplainer on the true network (rank-correlation of attributions reported). The fidelity number turns a weakness into a transparency feature. |

## 1.D Backend, frontend, deployment and academic-risk flaws

### F-26 [HIGH] Training on Railway; simulation inside the API event loop

| | |
|---|---|
| **v2 says** | One "api" container mounts ./models and carries WANDB_API_KEY; POST /simulate runs jobs via "Celery/BackgroundTasks" in the same service; Railway hosts everything. |
| **Why it is a problem** | Railway containers are CPU-small with ephemeral filesystems: training there is impossible and mounting ./models does not survive redeploys. A 70-lap, 22-car simulation running inside FastAPI BackgroundTasks blocks the event loop the WebSocket needs, so the live view stutters exactly when it is being demonstrated. |
| **Fix (v3)** | Training happens off-platform (local GPU / university HPC / Colab — OQ-2); the deployed system serves a frozen artifact baked into the worker image. Architecture: api (FastAPI) + worker (RQ consuming a Redis queue, running simulations and publishing lap events) + static frontend + managed Postgres/Redis. Section 5.11. |

### F-27 [HIGH] VITE_API_URL: the browser cannot reach http://api:8000

| | |
|---|---|
| **v2 says** | Compose sets VITE_API_URL=http://api:8000 and VITE_WS_URL=ws://api:8000 for the frontend container. |
| **Why it is a problem** | Vite env vars are baked into the JS bundle at build time and executed in the **user's browser**, which cannot resolve Docker-internal hostnames. Every API call in the deployed demo would fail with a DNS error — discovered, traditionally, five minutes before the demo. |
| **Fix (v3)** | Frontend uses relative /api and /ws paths behind a reverse proxy locally, and the public Railway API URL in production builds; wss:// in production (mixed-content rule). Section 5.11 environment table. |

### F-28 [MEDIUM] 22 per-car DOM labels and six live canvases will not hold 60 fps

| | |
|---|---|
| **v2 says** | Race Replay puts a Drei Html label on every car; each of the six views is "a separate R3F Canvas"; bloom runs everywhere; target is 60 fps on a GTX 1660. |
| **Why it is a problem** | Drei Html creates a real DOM node per label, reprojected every frame: 22 labels is hundreds of layout/style operations per frame, the single most common R3F performance killer. Multiple simultaneously mounted canvases each own a WebGL context (browsers cap ~8-16 and each costs memory). |
| **Fix (v3)** | One canvas mounted per route (views are pages, not tabs of live canvases); labels only for the selected car plus podium, others as instanced sprite text; cars via InstancedMesh with per-instance colour; bloom behind a quality toggle; perf budget table with measured checks in Section 5.10. |

### F-29 [MEDIUM] Team liveries and sponsor decals are a trademark problem

| | |
|---|---|
| **v2 says** | Recruiter Phase Month 6: "car liveries (team colours + sponsor decals)"; demo script narrates "Verstappen's car leads". |
| **Why it is a problem** | F1, team and sponsor marks are protected; a public portfolio site reproducing liveries and sponsor logos is exactly the artefact a team's legal-adjacent reviewer notices for the wrong reason. FastF1/OpenF1 are unofficial sources — fine for research, but the site must not look like an official product. |
| **Fix (v3)** | Generic liveries keyed by team colour only, no sponsor marks, an "unofficial project, not associated with Formula 1" footer (mirroring OpenF1's own disclaimer), and driver labels as three-letter codes from the data. R-07. |

### F-30 [MEDIUM] AI-drafted dissertation needs explicit policy compliance

| | |
|---|---|
| **v2 says** | Section 12: "AI-assisted dissertation... Generate full 15,000-word draft" via Claude. |
| **Why it is a problem** | University of Liverpool COMP702 has rules on generative-AI use and declaration; "AI drafted it, I reviewed it" sits differently under different departmental policies, and an undeclared mismatch is an academic-integrity risk that no technical quality can offset. |
| **Fix (v3)** | Action OQ-3: obtain the current COMP702 AI-use policy in writing and follow its declaration format. Default posture in v3: you write the dissertation; AI assists with editing, figures and code — and the assistance is declared. The engineering spec (this document) is unaffected. |

# Part 2 — Edge cases and failure modes (with causal mechanisms)

v2's edge-case table (Section 9.2) is a genuine strength and is carried forward. The cases below are the non-obvious ones it missed. For each: the mechanism — why it happens and what it silently corrupts — then the required handling. Grouped by layer.

## 2.1 Simulation and regulation edge cases

| ID | Failure mode | Causal mechanism — why it happens and what it hides | Required handling |
|---|---|---|---|
| E-01 | SOC depletion cascade | Deploying more than you harvest for consecutive laps drains the 4 MJ store; once SOC hits the floor the car derates on straights, loses places, falls into dirty air, brakes later behind traffic and harvests less — a self-reinforcing spiral. If the simulator lets SOC go negative or refills it magically, the agent never learns the real cost of aggression and every "energy banking" finding is an artefact. | SOC hard-clipped to [0, 4] MJ; requested deployment above available energy is physically derated with an explicit lap-time penalty term; property test: 100k random action sequences, SOC never leaves bounds, energy conserved to 1e-9 MJ. |
| E-02 | Per-circuit harvest caps that move mid-season | The FIA adjusts the recoverable-energy limit per event (baseline 8.5 MJ; roughly 5-9 MJ range; reduced further in some qualifying sessions). A constant 8.5 in code makes Monza strategy wrong by construction and dates the project the next time the FIA tweaks a value. | H_max is a per-event field in circuits.yaml with a source note and date; the assumptions register tracks it; the env reads config, never a literal. |
| E-03 | VSC treated as SC | A virtual safety car slows everyone by a delta with no bunching; a full SC bunches the field and roughly halves relative pit loss. Collapsing both into one flag (v2 state #16) makes the agent value VSC windows like SC windows — it will pit under VSC when staying out was right, and the "SC exploitation" metric silently mixes two different opportunities. | Separate track states (green / VSC / SC / red) one-hot in the observation; separate pit-loss multipliers fitted from 2022-2026 data; metrics reported per state. |
| E-04 | Red flag = free tyre change | Under red flag, cars may change tyres in the pit lane at zero race-time cost. This single rule can invert an entire strategy (track position becomes king). v2 models red flags but not the free change, so simulated red-flag races diverge wildly from the real races used for validation — and the divergence looks like agent error. | Red-flag handler grants a compound choice to all cars at restart; noted in era config (applies to both eras); validation excludes or era-matches red-flagged real races. |
| E-05 | Mass-pit ordering under SC | When the field pits on the same lap, exit order depends on pit-lane queueing and stationary time variance. Resolving simultaneous stops naively by pre-pit order deletes the very mechanism (double-stacking losses, pit-lane traffic) that makes SC decisions interesting. | Pit lane modelled as a FIFO with per-car service time draws; positions recomputed from cumulative race time, never swapped in place; property test with 22 simultaneous stops. |
| E-06 | Rival archetype degeneracy under SC | If all three scripted archetypes share one "pit under SC" rule, every SC produces the same field response; the agent memorises that pattern, and the 78% "SC exploitation" score measures pattern-matching against your own script — it can collapse against any other opponent behaviour. | Archetype parameters (pit thresholds, SC response probability, pace sigma) drawn per race from documented distributions; a held-out parameter band is reserved for evaluation only (train/test split on opponents), Section 5.12. |
| E-07 | Backmarker traffic after pitting | A pit stop drops the agent into slower traffic; if lapped cars are not modelled as moving obstacles with dirty air, the simulator prices pit stops too cheaply and the agent systematically over-pits — which then looks like "aggressive strategy discovery". | All 22 cars simulated with the same pace/traffic model; overtaking a slower car costs a fitted time penalty scaled by pace delta and override availability. |
| E-08 | Wet-transition cliff timing | Crossover between slicks/inters/wets is a threshold on continuous wetness; if wetness updates once per lap while the threshold sits mid-lap, the optimal switch lap is ambiguous by one lap — exactly the margin the British GP counterfactual is meant to test. | Wetness integrates sub-lap (3 sectors); compound validity evaluated at pit-entry time; the ambiguity is stated in the counterfactual write-up. |

## 2.2 Data pipeline edge cases

| ID | Failure mode | Causal mechanism | Required handling |
|---|---|---|---|
| E-09 | In/out laps poison tyre fits | PitIn/PitOut laps carry NaT or inflated LapTime; if they enter the degradation regression, every stint appears to end with a cliff, and the fitted "thermal runaway" is partly a data-cleaning bug wearing a physics costume. | Exclude in/out laps and IsAccurate==False laps from fits; report per-fit counts of excluded rows in the data-quality report. |
| E-10 | Track evolution vs fuel burn identifiability | Lap times fall through a race because fuel burns off AND the track rubbers in. Fitting either alone absorbs the other's effect; v2 fixes fuel at 0.035 s/kg and evolution at -0.04 s/lap independently, which double-counts the trend and biases tyre degradation the other way. | Joint fit per race: lap_time ~ tyre(age,compound) + a*fuel_mass + b*lap_number with compound fixed effects; report collinearity; carry the fitted a, b into the simulator. |
| E-11 | Sprint weekends break session assumptions | Sprint formats change session names/order and give fewer representative race laps; ingestion code that assumes FP1-FP2-FP3-Q-R crashes or mislabels, and per-circuit models quietly train on half the expected data. | Ingest by session type from the event schedule API, not by fixed name list; per-event row counts asserted in the quality report. |
| E-12 | Jolpica / OpenF1 rate limits during bulk ingest | Jolpica is a free community service with request budgets; OpenF1 free tier allows about 3 req/s. A naive per-lap loop across ~70 weekends trips 429s mid-ingest and leaves a partially-written database that later reads as "2023 had fewer pit stops". | One-shot bulk ingest with FastF1 on-disk cache, client-side throttling, resumable checkpoints per event, and a completeness assertion (expected vs loaded events) before any modelling. |
| E-13 | Madrid 2026 has no data until September | A brand-new street circuit joins the calendar in September 2026; any "all circuits" claim or hard-coded list breaks, and the ribbon pipeline has no GPS source for it until FP1. | Circuit library is generated from ingested data (F-08); Madrid auto-appears after its first session; claims phrased as counts computed at build time. |
| E-14 | 2026 rounds as both calibration and test set | Using all available 2026 races to tune the synthetic generator and then "validating" on the same races is train-test contamination — the validation chapter would be circular. | Freeze the split now: rounds 1-6 calibrate, later rounds (and all future rounds as they happen) are held-out test. The split date is recorded in the repo. |

## 2.3 Serving and frontend edge cases

| ID | Failure mode | Causal mechanism | Required handling |
|---|---|---|---|
| E-15 | Redis cache serving a stale model's results | Completed simulations are cached; after retraining, identical request bodies hit the old cache and the demo shows decisions the new model would not make — an explainability nightmare, since the SHAP panel then "explains" a ghost. | Cache keys include model_version and env_config_hash; deploying a model bumps the version and cold-starts its own namespace; TTL 7 days as backstop. |
| E-16 | WebSocket consumer slower than producer | The worker can simulate 70 laps in under a second; pushing events at production speed floods the socket and the browser animation timeline. Naive sleeps in the worker then couple simulation speed to playback speed. | Worker publishes the complete lap log to Redis/Postgres immediately; the API streams to clients on a playback clock (1x/5x/10x/30x) from the stored log — pause/scrub/reconnect all read the same source of truth. v2's reconnect requirement is kept. |
| E-17 | Ribbon self-intersection at hairpins | Extruding a 12 m ribbon along a curve whose radius is under 6 m makes inner edges cross; Frenet frames also flip at inflection points, twisting the mesh. Monaco and Hungary produce visibly broken geometry precisely on the corners recruiters zoom into. | Rotation-minimizing frames; clamp effective half-width to local radius; geometry unit test asserts no self-intersection across all circuits (E-17 test in Section 5.10). |
| E-18 | GPU context loss on mobile / long sessions | Browsers reclaim WebGL contexts under memory pressure; a black canvas mid-demo with no handler looks like a crash. | Listen for webglcontextlost / restored; show a reload overlay; keep one canvas per route (F-28) to minimise pressure. |

# Part 3 — Removed from v2 (and exactly why)

Everything here was cut because it would create a problem later — a wrong claim, a dead dependency, an impossible compute bill, or a legal exposure. Each row names the finding that justifies it.

| ID | Removed item | Why it would have hurt you later | Basis |
|---|---|---|---|
| R-01 | Action 11 "push for fastest lap" + its reward term | Optimises a championship point abolished from 2025; indefensible in a 2026 project. | F-07 |
| R-02 | Reward terms: SC-pit bonus, ERS-efficiency, grip-at-stint-end, undercut bonus, position +1.0/-0.8 asymmetry, ERS violation penalty, fuel penalty, tyre-failure -12 | Reward hacking (position pump), circularity (paying for the "discoveries"), and perverse incentives (early pitting). Twelve terms collapse to three; behaviours become metrics. | F-15, F-16, F-17 |
| R-03 | The 4 MJ/lap deployment-cap mechanic, its -8 penalty, its red cap plane and its "99.5% compliance" metric | Built on a rule that is not in the 2026 rulebook. | F-02 |
| R-04 | Ergast API dependency | Service shut down at the start of 2025; calls fail today. | F-10 |
| R-05 | OpenF1 real-time features and any 2022 OpenF1 usage | Real-time stream is paid; 2022 not covered. The live view runs on your own simulator. | F-11 |
| R-06 | DQN ablation; ablation budget shrunk 8 runs to 4 (+1 optional) | SB3 DQN cannot consume the MultiDiscrete action space; 160M extra steps is not a one-student compute budget. | F-18, F-22 |
| R-07 | Team liveries, sponsor decals, real-driver framing on the public site | Trademark exposure on the exact artefact you want hiring teams to inspect. | F-29 |
| R-08 | Mobile 3D and broadcast-camera work inside the core build | Pure polish competing with evaluation time; moved to the optional recruiter phase. | Scope (F-26, Part 6) |
| R-09 | GAN mention for synthetic data | Unused, unbuilt, and invites "why not?" questions; physics scaling + real-2026 calibration is the defensible route. | Part 5.4 |
| R-10 | 8 MJ battery and all 0-8 MJ ranges | Wrong capacity removes the scarcity that creates the strategy problem. | F-03 |
| R-11 | Observation features #22 (target position delta) and #7 (ERS deployed this lap) | Objective leakage into the state; a within-lap quantity at lap granularity. | F-19 |
| R-12 | 20M-step default training budget | Unpriced compute risk; replaced by an 8M default with competence gates and an extension option once throughput is measured. | F-21, F-22, Section 5.6 |

# Part 4 — Added in v3 (and what each addition buys)

| ID | Addition | What it buys you | Where |
|---|---|---|---|
| A-01 | Action masking via MaskablePPO (sb3-contrib) | 100% regulatory legality by construction — a stronger claim than any compliance percentage, and cleaner learning. | 5.5, 5.6 |
| A-02 | Sim-to-real calibration + validation against real 2026 rounds (fixed 1-6 / 7+ split) | Converts the speculative synthetic-2026 chapter into measured evidence; the single biggest credibility upgrade available to this project. | 5.4, 5.12 |
| A-03 | Decision cards: top-5 SHAP + counterfactual flip point per decision | A recruiter-readable, mechanically honest explanation of every pit call. | 5.8 |
| A-04 | Surrogate distillation with reported fidelity (+ gradient cross-check) | Makes SHAP tractable and turns the approximation into a transparency feature. | 5.8 |
| A-05 | Oracle-regret metric (dynamic-programming strategist on the deterministic env) | Answers "good compared to what ceiling?" — the question position-gain metrics dodge. | 5.12 |
| A-06 | Paired-seed evaluation with bootstrap confidence intervals | Cancels safety-car luck between agent and baseline; every headline number ships with an interval. | 5.12 |

| ID | Addition | What it buys you | Where |
|---|---|---|---|
| A-07 | Three scripted baselines (engineer heuristics / replayed real strategy / random-legal) | Position gains become "vs named baseline under identical randomness", not "vs the universe". | 5.12 |
| A-08 | Safety-car lottery variance decomposition (formalised from v2's idea list) | A genuinely novel analytics result: how much finishing position is luck vs decision quality. | 5.12 |
| A-09 | Reproducibility pack: seed manifest, config hashes, data snapshot, one-command Makefile | Examiner can rerun the headline table; hiring managers notice. | 5.6, 5.11 |
| A-10 | Era config (2024 vs 2026 constants) for era-matched counterfactuals | Separates "better strategy" from "different car". | 5.5, F-24 |
| A-11 | Interpretability contract (Part 0.4) enforced per component | Your non-negotiable requirement, made auditable. | 0.4 |
| A-12 | MoSCoW scope ladder + Minimum Defensible Product gate at week 6 | Guarantees a submittable dissertation even in the worst timeline. | Part 6 |

## 4.1 What v2 got right — carried forward unchanged

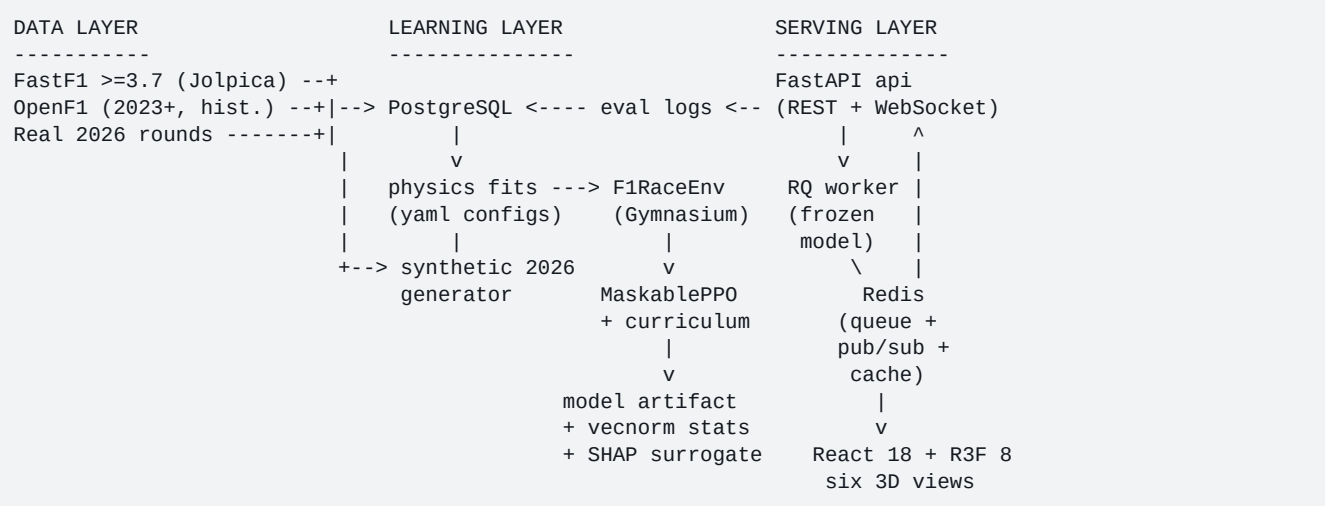
Credit where due; these survive intact because they are correct and well-judged:

- **Lap-level decision granularity.** One decision per lap keeps episodes at 44-70 steps — the right horizon for PPO and the right abstraction for strategy (engineers think in laps).
- **Physics as separate, unit-tested modules** (tyre / energy / fuel / aero / events) assembled by the environment. This decomposition is exactly how it should be built.
- **The six-view dashboard concept**, especially the split-viewport counterfactual view — the single most persuasive recruiter artefact in the design.
- **The WebSocket reliability requirement** (reconnect + replay-from-store). Kept verbatim and strengthened by E-16.
- **The unit-test-first table** (Section 9.1) — extended, not replaced.
- **W&B experiment tracking** and checkpointing discipline.
- **The 10 counterfactual race choices** — a well-curated strategic test set (now run era-matched per F-24).
- **gamma = 0.997 reasoning** — the 0.997^69 = 0.81 argument is correct and stays, with a sensitivity note.
- **Repository structure** (Section 3.2) — sensible; only a worker/ service and configs/ directory are added.
- **The recruiter demo script** — strong storyboard; updated only where mechanics changed (energy view, liveries, live data).

# Part 5 — The rebuilt master specification

Everything from here down is the build contract. It is written so that each numbered section can be pasted into Claude Code as the context for one work package, together with the matching tests. No magic numbers: every constant is either sourced (Part 8), fitted from data (with the fitting procedure stated), or explicitly marked as a stylised choice.

## 5.1 System architecture




Three invariants: (1) training never happens in the serving layer; (2) every number the frontend displays is fetched from the API, which reads Postgres — no client-side physics; (3) every model artifact ships as a triplet: policy weights + VecNormalize statistics + SHAP surrogate, versioned together.

## 5.2 Corrected 2026 physics model (the constants that drive everything)

These replace v2 Section 4.5. Column "Conf." is the confidence grade from the Assumptions Register (Part 8): A = verified against two or more primary sources; B = single good source; C = estimate to be fitted or verified before use.

| Quantity | 2024 era | 2026 era | Source / note | Conf. |
|---|---|---|---|---|
| MGU-K max power | 120 kW | 350 kW | FIA 2026 PU regs Art. 5; Formula1.com explainers | A |
| ICE power (approx.) | 550-560 kW | ~400 kW | Formula1.com / ESPN 2026 guides | A |
| Deployment speed taper | n/a | full 350 kW to 290 km/h, linear to 0 at 355 km/h | Formula1.com 2026 PU explainer | A |
| Manual Override (replaces DRS as overtaking aid) | DRS flap | 350 kW held to 337 km/h, ~0.5 MJ extra, within 1.0 s at detection | Formula1.com; ESPN | A |
| Energy Store usable capacity | 4 MJ | 4 MJ (unchanged) | ESPN 2026 rules guide; FIA Art. 5.4 (cross-check) | A |
| Harvest cap per lap | ~2 MJ (MGU-K) | 8.5 MJ baseline; FIA-adjusted ~5-9 MJ per event; may be lower in qualifying | FIA regs + Raceteq / Autosport 2026 reporting | A |
| Harvest mechanisms | braking (+MGU-H) | braking; lift-and-coast at 350 kW (disables X-mode); super clipping at ~250 kW on throttle | Raceteq May 2026; Autosport Mar 2026 | B |
| MGU-H | present | abolished | FIA 2026 regs | A |
| Active aero | none (DRS only) | X-mode (low drag, designated sections) / Z-mode (high downforce); available to all cars | Formula1.com aero explainer | A |
| Downforce / drag vs 2024 | baseline | downforce about -30%, drag about -55% | Honda 2026 regulations overview | B |
| Minimum car mass | ~798 kg | 768 kg | Honda 2026 overview (-30 kg) | A |

| Quantity | 2024 era | 2026 era | Source / note | Conf. |
|---|---|---|---|---|
| Race fuel mass | ~100 kg | ~70 kg (energy-flow limited, 100% sustainable) | VERIFY vs FIA 2026 regs before Phase 2 | C |
| Fastest-lap point | existed to 2024 | abolished from 2025 | VERIFY vs 2026 Sporting Regs (30-second check) | B |
| Grid size | 20 cars | 22 cars (11 teams incl. Cadillac) | 2026 season fact | A |
| 2026 calendar | n/a | 22 rounds after two cancellations; Madrid new in September; ~10 rounds raced by 7 July 2026 | Sky Sports / season reporting | A |

*Design consequence: the strategic tension of 2026 is a 4 MJ tank feeding a 350 kW motor, refilled only by spending lap time. Every layer of v3 — actions, observation, physics, visualisation, metrics — is organised around SOC scarcity rather than the fictional per-lap cap.*

## 5.3 Data pipeline — exact sources, columns and cleaning contract

### 5.3.1 Sources and versions

| Source | Pinned | Covers | Used for | Limits to respect |
|---|---|---|---|---|
| FastF1 (Jolpica backend) | fastf1>=3.7,<4 | 2022-2026 sessions: laps, car/pos telemetry, weather, race control, results, schedules; circuit corner info | Everything: physics fits, replays, ribbons, results | Enable on-disk cache before any call; bulk ingest once; Jolpica request budget respected via cache |
| OpenF1 REST | HTTP v1 | 2023-2026: pit durations, stints, intervals, positions, race control, weather | Pit-loss distributions per circuit; interval cross-checks | Free tier ~3 req/s; historical only (real-time is paid); no 2022 |
| Real 2026 rounds | via both above | Rounds raced so far (~10 by 7 July) | Calibration (rounds 1-6) and held-out validation (rounds 7+) | Split frozen now (E-14) |

### 5.3.2 FastF1 objects and the columns actually consumed

| Call | Columns consumed | Feeds |
|---|---|---|
| fastf1.get_event_schedule(year) | RoundNumber, EventName, EventFormat, Session dates | Ingest plan; sprint handling (E-11) |
| session.laps | Driver, DriverNumber, Team, LapNumber, LapTime, Stint, Compound, TyreLife, FreshTyre, PitInTime, PitOutTime, Sector1/2/3Time, SpeedST, Position, TrackStatus, IsAccurate, LapStartTime | Tyre degradation fits; pit timing baselines; position traces; track-status timeline |
| lap.get_car_data() | Speed, Throttle, Brake, nGear, RPM, DRS at roughly 4-5 Hz (about 240 ms) — not 10 Hz as v2 stated | Braking-energy integrals; full-throttle fraction for deployable-energy estimates |
| lap.get_pos_data() | X, Y, Z (decimetres), Status | Circuit ribbon centreline (F-12); replay splines |
| session.weather_data | AirTemp, TrackTemp, Humidity, Rainfall, WindSpeed (per minute) | Weather state machine; temperature modifiers |
| session.race_control_messages | Category, Flag, Status, Message, Lap | SC/VSC/red timelines; per-circuit event rates |
| session.results | Abbreviation, TeamName, GridPosition, Position, Status, Points | Baselines; counterfactual ground truth; rival pace priors |
| fastf1.get_circuit_info() | corners (Number, Distance, X, Y, Angle), rotation | Corner annotations draped on the ribbon (not the centreline itself) |

### 5.3.3 OpenF1 endpoints (historical only)

| Endpoint | Key params | Fields consumed | Feeds |
|---|---|---|---|
| /v1/sessions | year, session_name | session_key, meeting_key | Key mapping for all other calls |
| /v1/pit | session_key | pit_duration, lap_number, driver_number | Pit-lane loss distribution per circuit (mean + sigma for the FIFO model, E-05) |
| /v1/stints | session_key | compound, lap_start, lap_end, tyre_age_at_start | Cross-check of FastF1 stints; 2026 compound behaviour |
| /v1/intervals | session_key | gap_to_leader, interval | Gap dynamics sanity checks |
| /v1/race_control | session_key | category, flag, message | SC/VSC ground truth cross-check |
| /v1/weather | meeting_key | air/track temperature, rainfall | Weather calibration 2023+ |

### 5.3.4 Cleaning contract (applied before any model fit)

- Drop laps with IsAccurate == False; drop in-laps and out-laps (PitInTime or PitOutTime set) from pace and degradation fits — they remain in the pit-timing dataset.
- Parse TrackStatus digit strings; a lap is "green" only if the whole lap is status 1; SC/VSC laps are excluded from pace fits and included in the event-rate fits.
- LapTime NaT: never impute for fits; impute only for replay display, flagged in the row.
- Telemetry gaps over 2 s: split the trace; never interpolate through a gap for energy integrals.
- Position data: convert decimetres to metres; drop points with Status != "OnTrack"; per-session z-score outlier gate on point-to-point speed.
- Wet or red-flagged races are excluded from dry tyre fits and analysed separately.
- Per season, emit data_quality_report.md: events expected vs loaded, rows kept vs dropped per rule, channels missing per session. Modelling may not start until the report exists (F-14).

### 5.3.5 Storage

The v2 schema (Section 8.3) is kept with these changes: laps gains energy_deployed_mj, energy_harvested_mj, soc_end_mj (simulated rows only) and an era + source flag (real_2022 ... real_2026 / synthetic_2026); agent_decisions stores one JSONB per decision (state, masks with reasons, action, probs, shap top-k) rather than one row per feature; simulation_runs gains env_config_hash and model_version (E-15).

## 5.4 Synthetic 2026 generation — eight named transformations

Purpose: produce 2026-era race dynamics for every circuit, including those the 2026 season has not visited yet. Method: transform fitted 2024 behaviour with sourced physics deltas, then calibrate the residual against real 2026 rounds 1-6 and validate on rounds 7+ (A-02, E-14). "No GAN" stands: every transformation is a formula you can defend line by line.

| ID | Transformation | Formula / procedure | Rationale and source |
|---|---|---|---|
| T1 | Powertrain reprofile | Recompute straight-line speed profiles with a point-mass longitudinal model: P(v) = P_ICE + P_K(v) with P_K = 350 kW to 290 km/h tapering to 0 at 355; drag from T2. Output per circuit: full-throttle time, terminal speeds, deployable energy integral E_dep. | FIA power/taper figures (5.2). This is what converts telemetry into 2026 energy budgets. |
| T2 | Aero scaling | Straights: drag force x0.45 in X-mode sections (source: about -55% drag). Corners: apparent grip from downforce x0.70 passed through the corner-speed model (v_corner scales roughly with the fourth root of downforce at fixed radius; fitted, not assumed). | Honda 2026 overview; fitted exponent documented in the calibration notebook. |
| T3 | Mass shift | Minimum mass 768 kg; lap-time mass sensitivity k_m fitted per circuit from the joint fuel/evolution regression (E-10), applied to the -30 kg delta and the new fuel profile. | Honda 2026 overview; avoids v2's fixed 0.035 s/kg. |

| ID | Transformation | Formula / procedure | Rationale and source |
|---|---|---|---|
| T4 | Energy accounting layer | Per circuit: H_max from config (default 8.5 MJ); harvest-by-mode = braking integral + mode-dependent lift-and-coast term with pace cost c_lc (s/MJ) and clipping term; E_dep from T1. These four numbers per circuit are the entire energy game. | F-02/F-04 corrections; Raceteq/Autosport mechanism reporting. |
| T5 | Tyre compounds 2026 | Refit degradation curves grip(age) = 1 - a*age^b per compound on real 2026 rounds 1-6; for circuits without 2026 data, transfer coefficients from the nearest cluster (clustering on 2022-2024 abrasiveness + layout features), flagged low-confidence. | Pirelli 2026 compounds differ; measured beats guessed. |
| T6 | Fuel/energy budget | Initial fuel mass from the race energy allowance (Conf. C — verify); burn profile scaled so total burn = allowance; weight effect through T3. | F-06 correction. |
| T7 | Overtaking model | Pass probability P(pass \| pace delta, override available) refit on 2026 rounds 1-6 overtake counts; prior: override is worth roughly 0.1-0.2 s (vs DRS 0.6-0.8 s), so passing is harder than 2024. | ESPN 2026 guide; fitted, then checked against round 7+ overtake counts. |
| T8 | Residual recalibration | After T1-T7, per calibration circuit fit the leftover offset c_circuit = median(real 2026 lap time) - median(simulated); apply; report held-out error on rounds 7+ as the headline sim-to-real number. | Standard model-correction step; keeps all physics terms interpretable while absorbing unmodelled effects into one named constant. |

### 5.4.1 Validation targets for the generator (reported in Chapter 6)

- Lap-time distribution: Kolmogorov-Smirnov distance between simulated and real 2026 green-lap distributions per held-out circuit; target KS under 0.15, reported per circuit.
- Stint lengths: simulated vs real stint-length histograms per compound; report earth-mover distance and comment on outliers.
- Pit loss: simulated pit delta within 1.0 s of the OpenF1-measured mean per circuit.
- Event rates: SC/VSC per race within the 2022-2026 binomial confidence band per circuit.
- Energy plausibility: simulated derate frequency qualitatively consistent with 2026 broadcast/reporting (cars running dry on straights at energy-hard circuits).

## 5.5 Gymnasium environment contract (F1RaceEnv)

One environment class, parameterised by era (2024 or 2026 constants file) and circuit config. Decision frequency: one step per lap (kept from v2 — see Part 4.1). Episode length: the race distance, 44-70 steps.

### 5.5.1 Observation space — Box(low=-1, high=1, shape=(32,), float32)

Every dimension named, bounded and justified. VecNormalize runs on top during training; the raw encodings below are already scale-bounded so the served model degrades gracefully if statistics drift.

| # | Name | Encoding / range | Why the strategist needs it |
|---|---|---|---|
| 1 | lap_progress | lap / total_laps, [0,1] | Race phase changes the value of every action; laps_remaining is its complement so one scalar suffices |
| 2 | position_norm | (position - 1) / 21, [0,1] | Objective context; 22-car grid |
| 3 | grid_delta | (grid - position) / 21, [-1,1] | Whether the race is being won or lost vs start — cheap trend signal |
| 4-8 | compound one-hot | S / M / H / Inter / Wet | Degradation anchor (kept from v2) |
| 9 | tyre_age_norm | age / 40, clipped [0,1] | Primary degradation driver |
| 10 | grip_est | model grip estimate, [0,1] | The simulator's own tyre state — the analogue of an engineer's deg model |
| 11 | tyre_temp_state | {cold 0, optimal 0.5, hot 1} | Cliff-risk signal |
| 12 | pit_count_norm | stops / 3, clipped [0,1] | Strategy shape so far |
| 13 | second_compound_used | {0,1} | The mandatory two-compound rule (F-20) — without this the rule is invisible to the policy |
| 14 | fuel_frac | mass / initial, [0,1] | Weight effect and burn planning |
| 15 | soc_frac | SOC / 4 MJ, [0,1] | The central 2026 resource |
| 16 | last_lap_harvest_frac | harvested / H_max, [0,1] | Energy-economy feedback: is banking working |
| 17 | last_lap_deploy_frac | deployed / E_dep, [0,1] | Deployment feedback; replaces v2's within-lap #7 (F-19) |
| 18 | gap_ahead_norm | min(gap, 30 s) / 30 | Undercut / attack signal |
| 19 | gap_behind_norm | min(gap, 30 s) / 30 | Defence signal |
| 20 | ahead_tyre_age_delta | (their age - mine) / 40, [-1,1] | The quantity real pit walls stare at before an undercut |
| 21 | ahead_compound_delta | hardness(theirs) - hardness(mine), scaled [-1,1] | Pace-vs-longevity context for the car ahead |
| 22 | behind_tyre_age_delta | as #20 for the car behind | Anti-undercut context |
| 23 | override_available | {0,1}: gap ahead under 1.0 s | The 2026 overtaking aid (F-05); replaces v2's "active aero available" |
| 24-26 | track_status one-hot | green / VSC / SC | VSC and SC are different opportunities (E-03); red flag is handled as an event, not a state the policy acts in |
| 27 | pit_open | {0,1} | Pit-entry legality context (also enforced by mask) |
| 28-30 | weather one-hot | dry / damp / wet | Compound validity regime |
| 31 | wetness | continuous [0,1] | Crossover timing lives on this scalar (E-08) |
| 32 | track_temp_norm | (T - 15) / 45, clipped [0,1] | Degradation modifier |

*Removed vs v2: #22 target-position-delta (objective leakage) and #7 ERS-deployed-this-lap (within-lap quantity) — see F-19. Rival context is now continuous tyre information about the actual neighbours instead of three binary archetype flags.*

### 5.5.2 Action space — MultiDiscrete([6, 4]) with masking

| Head | Values | Mask rules (each rule logged with a reason code) |
|---|---|---|
| A: pit decision | 0 stay out · 1 pit Soft · 2 pit Medium · 3 pit Hard · 4 pit Intermediate · 5 pit Wet | Pit choices masked while pit lane closed; slicks masked when wetness above the slick threshold; Inter masked outside damp band; Wet masked unless wet; stay-out masked on the last feasible pit lap of a dry race if second_compound_used == 0 (F-20) |
| B: energy / pace mode | 0 conserve (lift-and-coast harvest, pays pace) · 1 balanced · 2 attack (max deploy toward E_dep) · 3 defend (deployment weighted to the car behind) | attack masked when SOC below 0.5 MJ (nothing to deploy — masking beats letting the agent learn a no-op); defend meaningful only with a car within 2.0 s behind but never illegal, so unmasked |

Legality that is continuous rather than discrete (energy conservation, SOC bounds, harvest caps) is enforced inside the dynamics by clipping and derating — the agent can request, physics disposes. Together with the masks this is what makes "100% compliant by construction" true (F-16), and the 100k-random-action property test proves it.

### 5.5.3 Transition dynamics (per lap, per car — all 22 cars run the same model)

```
lap_time = t0(circuit, era)                    # fitted per circuit per era
         + k_m * fuel_mass                     # k_m fitted jointly (E-10)
         + deg(compound, age, track_temp)      # grip model, T5
         + mode_pace(mode)                     # conserve +c_lc; attack -c_atk (energy-bounded)
         + traffic(gap_ahead, era)             # dirty air under 2.0 s, era-scaled
         + status_mult(track_status)           # VSC / SC multipliers, fitted
         + wet_penalty(wetness, compound)      # crossover model, E-08
         + eps,  eps ~ N(0, sigma_driver)      # per-car pace noise, fitted

harvest = min(H_max(circuit),
              H_brake(circuit) + [mode==conserve] * H_lc + H_clip(circuit))
deploy  = min(D_request(mode), E_dep(circuit), SOC + intra_lap_share * harvest)
derate_penalty = p_der * max(0, D_request(mode) - deploy)   # s per MJ shortfall
SOC'    = clip(SOC + harvest - deploy, 0, 4.0)

events: SC ~ hazard lambda_c per lap; VSC from per-car failure hazard;
        weather Markov per circuit; tyre-failure hazard beyond cliff age -> forced pit
        (+25 s, kept from v2); red flag -> positions per rules + free tyre choice (E-04)
ordering: cumulative race time; on-track passes gated by P(pass) (T7);
          pit lane is a FIFO with per-car service-time draws (E-05)
```

Every named constant above lives in physics_params.yaml with a comment stating its source: fitted (which notebook), sourced (which document, see Part 8), or stylised (explicitly labelled). That file is the anti-black-box artefact for the physics layer.

### 5.5.4 Reward — three terms, each with its coefficient argued

| Term | Formula | Coefficient and why exactly that value |
|---|---|---|
| Pace shaping | r_pace = w_p * (t_ref(c) - lap_time) / s_ref(c) | w_p = 0.02. s_ref is the per-circuit clean-lap standard deviation, making the term scale-free across circuits; over a 60-lap race its total magnitude is about 1.2 at most — an order below the terminal term, so it provides per-lap gradient without competing with the objective. Ablation 4 trains without it. |
| Position shaping (potential-based) | F = gamma * Phi(s') - Phi(s), with Phi(s) = -0.1 * position | Coefficient 0.1: a net one-place gain contributes +0.1 in total because the sum telescopes — its job is direction, not magnitude. Potential-based form (Ng, Harada, Russell 1999) provably leaves the optimal policy unchanged and pays exactly zero for the oscillation exploit that broke v2 (F-15). |
| Terminal points | R_T = points_2026(final position) / 2.5 (P1 = 10.0, P2 = 7.2, ... P10 = 0.4, else 0; DNF = 0) | Divisor 2.5 sets P1 = 10, about 3x the maximum cumulative shaping (~1.2 + ~2.1), so the terminal objective dominates — the dominance ratio is logged per run. Using the real points table makes the objective legible to any F1 person. DNF pays 0 rather than negative: retirement already forfeits everything; stacking an extra penalty double-counts risk (documented choice, optional ablation). |

Per-lap reward decomposition (all three terms separately) is written to the episode log for every evaluation race — the interpretability contract row for this component.

### 5.5.5 Termination, truncation and the info dict

- terminated = True at race distance, or on retirement (fuel/energy exhaustion, terminal car failure event). Tyre failure is NOT termination: forced pit + 25 s (kept from v2).
- truncated is never used: the horizon is the race by definition.
- Red flag: not a termination — field neutralised, positions per regulations, free tyre choice granted (E-04), race resumes.
- info per step: position, gaps, per-term reward decomposition, energy triplet (harvest, deploy, SOC), mask bit-vector with reason codes, event flags. info at episode end: final position, points, pit laps and compounds, derate count, energy totals.
- reset(seed=...) fully seeds every stochastic stream (events, rivals, noise) — the SubprocVecEnv statelessness requirement from v2 Section 13.2 is kept verbatim.

## 5.6 Training pipeline — MaskablePPO

### 5.6.1 Hyperparameters (default · range · reasoning)

| Parameter | Default | Range | Reasoning |
|---|---|---|---|
| Algorithm | MaskablePPO (sb3-contrib) | — | PPO stability on long horizons plus native MultiDiscrete action masking; the masking is the compliance architecture (F-16) |
| Network | MLP pi=[256,256], vf=[256,256] | [128,128] to [512,256] | 32-dim input needs no more; a small policy keeps surrogate distillation faithful (5.8) |
| n_envs | 12 (SubprocVecEnv) | 8-16 | One per physical CPU core; the env is pure Python + NumPy, so CPU parallelism is the throughput lever, not the GPU |
| n_steps | 512 per env | 256-1024 | Rollout of 6144 steps spans roughly 100 episodes — enough full races per update for stable advantage estimates on a 44-70 step horizon |
| batch_size | 1024 | 512-2048 | Divides the 6144 rollout evenly (6 minibatches) |
| n_epochs | 8 | 4-10 | Standard PPO reuse; watch approx_kl in W&B |
| learning rate | 3e-4, linear to 0 | 1e-4 to 5e-4 | SB3 default region; linear decay replaces v2 cosine — one fewer shape to justify |
| gamma | 0.997 | 0.995-0.999 | v2's argument kept: 0.997^69 = 0.81 preserves lap-1 influence at lap 70; sensitivity run at 0.999 noted |
| gae_lambda | 0.95 | 0.9-0.98 | Standard bias-variance point |
| clip_range | 0.2 | fixed | PPO standard |
| ent_coef | 0.01, linear to 0.002 | — | Explore early, commit late (kept from v2, endpoint raised slightly to avoid premature collapse in Stage C stochasticity) |
| vf_coef / max_grad_norm | 0.5 / 0.5 | fixed | SB3 defaults; no evidence they need touching here |
| VecNormalize | obs=True, reward=False | — | Observations vary in scale; reward is already hand-scaled and a running reward normaliser would shrink the sparse terminal signal. Statistics saved with every checkpoint (F-23) |
| Total budget | 8M steps (A 1.5M / B 2.5M / C 4M ceilings) | extend to 15M if throughput allows | Replaces v2's unpriced 20M (R-12); gates can end stages early |
| Seeds | 3 for headline, 2 per ablation | — | Mean and spread reported; single-seed RL results are noise |

### 5.6.2 Compute reality check (do this before believing any schedule)

Phase-2 exit includes a measured throughput number: steps/second across 12 workers for 100k steps. Expectation for a NumPy lap-level env with 22 cars: roughly 1,000-4,000 steps/s aggregate, putting 8M steps at 0.5-2.5 hours — training is then cheap and the 3-seed and ablation plans are safe. If measured throughput is below ~300 steps/s, profile the env (usual culprits: per-step Python object churn, pandas in the hot loop) before touching the budget. GPU

is optional: the policy is tiny; answer OQ-2 anyway for headroom.

### 5.6.3 Ablation set (4 runs + 1 optional, 4M steps and 2 seeds each)

| # | Run | Question it answers |
|---|---|---|
| 1 | No curriculum (Stage C from scratch) | Does staging actually buy sample efficiency or final quality here? |
| 2 | Penalty-based constraints instead of masks (v2's -8 architecture) | Direct test of F-16: quantifies violations during learning and any final-policy gap |
| 3 | v2's 12-term reward vs the v3 3-term reward | Is heavy shaping helpful or harmful? A likely publishable negative result on reward over-engineering |
| 4 | Terminal-only reward (both shaping terms off) | Are the shaping terms earning their place? |
| 5 (opt.) | A2C at matched budget | Algorithm sanity baseline if compute allows; DQN excluded with the documented MultiDiscrete reason (F-18, F-22) |

## 5.7 Curriculum — three stages with competence gates

| Stage | Environment | Exit gate (evaluated on frozen seed sets) | Why this gate |
|---|---|---|---|
| A — solo fundamentals | One circuit (Monza), dry, deterministic weather, no rivals, full 2026 energy model | On 100 fixed seeds: (i) eval return improvement under 1% across three consecutive evals (plateau) AND (ii) mean race time within 1.5% of the scripted solo reference (grid search over 1-3 stop plans x greedy energy schedule) | A solo race has a computable near-optimal reference, so competence is measured against an objective bar, not against the agent's own history |
| B — competition | Six circuits (Monza, Barcelona, Silverstone, Spa, Suzuka, Monaco — kept from v2, OQ-7), 21 rivals with mixed archetypes, dry, SC/VSC on | On 200 paired seeds vs the rule-based baseline in identical stochastic draws: finish-ahead rate at least 55% AND median position delta at least 0 | Paired seeds cancel safety-car luck, so the gate isolates decision quality (A-06); 55% requires real superiority beyond noise |
| C — full problem | All circuits with validated data, weather transitions, red flags, stochastic rival parameter draws (train band only, E-06), era 2026 | Stop when eval mean points stagnates for 1M steps; select the checkpoint by validation-seed points; report only on held-out test seeds, held-out rival band and held-out 2026 rounds | Standard train/validation/test discipline, applied to RL evaluation |

Step budgets from 5.6.1 are ceilings, not schedules (F-21). Gate values are logged to curriculum_gates.json — the interpretability contract row for this component. VecNormalize statistics carry across stages (a documented choice: continuity beats re-warming for an observation vector whose bounds do not change).

## 5.8 Explainability layer — what is explained, how, and how to say it

### 5.8.1 What a SHAP value means here (the mechanistic statement)

For one decision at one lap: the policy maps the 32-dimension state to a preference score (logit) for each action. The SHAP value of feature i is that feature's average marginal contribution to the chosen action's score, relative to a background distribution of typical race states, averaged over orderings (Shapley axioms). Positive pushes toward the chosen action; negative pushes away. It is an **attribution of the trained function's sensitivity** — it is not a causal claim about the world, and the write-up says so.

### 5.8.2 Pipeline (replaces v2 Section 6.5 — see F-25)

- **Distill:** log 200k (state, action-probability) pairs from evaluation rollouts; fit one gradient-boosted tree model per action head (XGBoost, multi-class) to imitate the policy.
- **Report fidelity:** top-1 agreement between surrogate and policy on held-out states — target at least 90% per head; also report mean KL. The fidelity number is printed on the dashboard next to every explanation. If fidelity misses 90%, deepen the surrogate before shipping explanations.

- **Explain:** exact TreeSHAP on the surrogate for every logged decision — fast enough for all evaluation races and for per-lap live cards (computed in the worker, streamed in the agent:decision event).
- **Cross-check:** GradientExplainer attributions on the true network for 500 sampled decisions; report the rank correlation with surrogate attributions as an integrity statistic in Chapter 7.
- **Background set:** 1,024 states stratified across circuits, race phases and track statuses; the background choice is documented because attributions depend on it.
- **Counterfactual flip probes:** for headline decisions, sweep one feature across its physical range with all else fixed and record where the policy's action flips ("stays out until tyre_age_norm = 0.42, i.e. about 17 laps"). Model-faithful, cheap, and the most persuasive line on a decision card.

### 5.8.3 The decision card (the recruiter-facing artefact)

| Field | Example content |
|---|---|
| Decision | Lap 28 - PIT for Medium + attack out-lap (probability 0.83) |
| Top drivers (TreeSHAP, fidelity 93%) | tyre grip est. -0.31 · gap behind 0.22 · ahead tyre age delta 0.19 · SOC 0.11 · lap progress 0.07 |
| Counterfactual flip | Would stay out if grip were above 0.58 (about 3 laps fresher) or if the car behind were more than 4.1 s back |
| Plain-language line | "Worn tyres plus an undercut threat from behind; the model banked energy last lap to attack the out-lap." |
| Honesty footer | Attribution of the trained policy via a 93%-faithful surrogate; not a causal claim. |

How to present it to a non-technical recruiter without losing accuracy: "For every decision, we show the five inputs that moved the model most, and the exact conditions under which it would have decided differently. The explanation comes from a simplified twin of the network that agrees with it 93% of the time — and that agreement number is always on screen." Every clause of that sentence is literally true of the pipeline above.

## 5.9 Technology stack (pinned) and FastAPI backend

### 5.9.1 Pinned stack — pin what breaks, float what does not

| Layer | Package | Pin | Why this pin |
|---|---|---|---|
| Runtime | Python | 3.11.x | Stable target for SB3 / FastF1 / SHAP together |
| Data | fastf1 | >=3.7,<4 | 3.7 is the Jolpica-backed line (F-10); major-pin against breaking changes |
| Data | pandas / numpy | >=2.2 / >=1.26 | FastF1 3.7 compatibility floor |
| RL | gymnasium | match SB3 pair | Version driven by the SB3 release used |
| RL | stable-baselines3 + sb3-contrib | same minor, e.g. 2.6.* | MaskablePPO lives in contrib; the pair must match minor-for-minor. Verify the exact latest pair at Phase 0 and freeze in the lockfile |
| RL | torch | >=2.2,<3 (CPU build fine) | Policy is tiny; CPU wheels simplify deployment images |
| XAI | shap / xgboost | >=0.45 / >=2.0 | TreeSHAP path of Section 5.8 |
| Backend | fastapi / uvicorn / pydantic | >=0.111 / latest / 2.x | v2 choices kept |
| Backend | SQLAlchemy / alembic / asyncpg | 2.0 / 1.13 / 0.29 | v2 choices kept |
| Backend | redis client / rq | 5.x / 1.16 | RQ is the simplest adequate job queue (F-26); Celery is overkill for one worker |
| Tracking | wandb | 0.17+ | Kept from v2 |
| Frontend | react + react-dom | 18.3 (NOT 19) | R3F 8 pairs with React 18; R3F 9 requires React 19 — mixing them is the most common R3F install failure |
| Frontend | @react-three/fiber / drei / three | ^8.17 / ^9.x / 0.16x exact pin | The triad must be mutually compatible; pin three exactly because drei helpers track specific three versions |
| Frontend | vite / tailwind / zustand / react-query / recharts / d3 | 5 / 3.4 / 4.5 / 5 / 2.12 / 7 | v2 choices kept |
| Infra | postgres / redis images | 16-alpine / 7-alpine | v2 choices kept |

*Lockfiles (pip-compile + package-lock.json) are committed in Phase 0; "pinned" means reproducible from the lockfile, not that these floors are the newest possible versions.*

### 5.9.2 REST endpoints (v2's map, corrected and trimmed)

| Method + path | Purpose | Response (Pydantic) |
|---|---|---|
| GET /api/v1/circuits | Circuit library with data-coverage flags (F-08) | CircuitSummary[] |
| GET /api/v1/circuits/{id} | Detail + GLTF asset URL + energy zone metadata | CircuitDetail |
| GET /api/v1/races/{year}/{round} | Ingested real-race data for replays | RaceDetail |
| POST /api/v1/simulations | Enqueue a simulation job (body below) | JobAccepted {run_id} |
| GET /api/v1/simulations/{run_id} | Status + summary result when complete | SimulationResult |
| GET /api/v1/simulations/{run_id}/frames?from_lap=&to_lap= | Paginated lap frames for replay/scrub (E-16 source of truth) | LapFrame[] |
| GET /api/v1/simulations/{run_id}/decisions | Decision cards incl. SHAP + flip probes | DecisionCard[] |
| POST /api/v1/counterfactuals | {year, round, driver} — era-matched replay (F-24) | JobAccepted |
| GET /api/v1/agents | Model registry: version, eval score, fidelity, config hash | ModelVersion[] |
| GET /api/v1/energy/{run_id} | Per-lap harvest/deploy/SOC series for the energy view | EnergyProfile |
| GET /api/v1/tyres/{circuit_id} | Fitted degradation surfaces per compound | DegradationCurve[] |
| GET /api/v1/health | Liveness for Railway health checks | {status, model_version} |

```
class SimulationCreate(BaseModel):
    circuit_id: str
```

```
    era: Literal["2024", "2026"] = "2026"
    grid_position: int = Field(ge=1, le=22)
    weather_profile: Literal["dry", "mixed", "wet_risk"] = "dry"
    rival_band: Literal["train", "test"] = "test"
    seed: int | None = None          # None -> random, echoed back for reproducibility
    model_version: str | None = None # None -> latest

class LapFrame(BaseModel):
    lap: int
    cars: list[CarState]             # 22 entries: id, s_track, position, compound,
                                     # tyre_age, gap_ahead_s, in_pit
    agent: AgentState                # soc_mj, harvest_mj, deploy_mj, mode, action,
                                     # reward_terms, mask_reasons
    events: list[RaceEvent]          # sc / vsc / red / weather / pit
```

### 5.9.3 WebSocket and worker

- WS /ws/simulations/{run_id}. Server events: sim:lap (LapFrame + DecisionCard), sim:event (SC/VSC/red/weather/pit), sim:complete (summary). Client events: client:speed {1|5|10|30}, client:pause, client:seek {lap}.
- The worker (RQ) runs the whole race immediately, persists every LapFrame and DecisionCard to Postgres, and publishes to Redis. The API streams to browsers on a playback clock reading the stored frames (E-16) — reconnection and scrubbing are therefore trivial, which keeps v2's "never lose a race to a dropped connection" requirement.
- Redis keys: cache sim:{model_version}:{env_hash}:{seed} (E-15); channel events:{run_id}; queue default RQ.

## 5.10 Frontend — six views, each with its assumptions stated

### 5.10.1 Circuit ribbon algorithm (replaces v2 Section 4.4 — see F-12, E-17)

- 1. Pick the fastest clean lap of the most recent dry session for the circuit; take get_pos_data() X, Y, Z; convert decimetres to metres.
- 2. Resample the centreline by arc length at 2 m spacing; close the loop; smooth X, Y and (separately, harder) Z with a Savitzky-Golay filter.
- 3. Build rotation-minimizing frames along the curve (parallel transport) — Frenet frames twist at inflection points and hairpins (E-17).
- 4. Extrude a 12 m ribbon; clamp half-width to local curve radius to prevent self-intersection; assert no self-intersection in the geometry unit test for every circuit.
- 5. Elevation: smoothed Z exaggerated x1.5, **labelled stylised** in the UI; where Z is missing, flat with corner camber — also labelled. Banking is a visual treatment, not data (F-12).
- 6. Annotate: corners from get_circuit_info(); braking zones from speed-drop detection on telemetry (drop above 50 km/h within 100 m — kept from v2); X-mode sections as the longest full-throttle stretches (stylised until official section data is available); pit lane as an offset spline.
- 7. Export GLTF + metadata JSON per circuit; lap length must match the official figure within 5% (kept from v2).

### 5.10.2 The six views: purpose, key change vs v2, and where each is brittle

| View | Kept purpose + change vs v2 | Brittleness / assumptions to state |
|---|---|---|
| 1 Circuit Explorer | As v2; "active aero zones" relabelled X-mode sections; energy annotations show harvest zones + per-circuit H_max | Ribbon is stylised in elevation and banking; X-mode sections are inferred, not official — both stated in the legend |
| 2 Race Replay | As v2 with 22 cars via one InstancedMesh (per-instance colour); labels only for selected car + top 3 (F-28); generic liveries (F-29) | Car positions between laps are spline-interpolated along track distance — mid-lap battles are illustrative, and the UI says so |
| 3 Energy View (was "ERS Landscape") | Rebuilt: SOC ribbon over laps + paired harvest/deploy bars against the per-circuit harvest cap line; the fictional 4 MJ deploy plane is gone (F-02) | Energy series are simulator-internal (F-13) — footer states "modelled, calibrated to 2026 aggregate behaviour" |

| View | Kept purpose + change vs v2 | Brittleness / assumptions to state |
|---|---|---|
| 4 Tyre Surface | As v2 (grip over age x temperature, vertex-shader surface, cliff visible) | Surface renders the fitted model, not raw data; compounds without 2026 fits show the transfer-cluster flag (T5) |
| 5 Live Simulation | As v2 but explicitly the agent in the simulator (R-05); decision card panel streams per lap with fidelity badge | Playback clock decouples sim speed from animation (E-16); WebGL context-loss overlay (E-18) |
| 6 Counterfactual | As v2 (split viewport, synced lap clock, divergence flash, delta chart) — era-matched runs (F-24) | Header states era config and "in-simulator comparison under stated assumptions"; the honesty is the credibility |

### 5.10.3 Performance budget (measured, not hoped)

| Budget item | Limit | Mechanism |
|---|---|---|
| Canvases mounted at once | 1 | Views are routes; navigating unmounts (F-28) |
| Draw calls in replay | under 150 | InstancedMesh for cars; merged ribbon geometry; LOD beyond 500 m (kept from v2) |
| DOM (Drei Html) labels | at most 4 | Selected car + podium; others use instanced sprite text |
| Postprocessing | bloom only, toggleable | Quality switch defaults off below a GPU heuristic; kept from v2's toggle idea |
| Frame rate | 60 fps target / 45 fps floor on a GTX 1660-class GPU | Chrome performance trace of a full 70-lap replay is a Phase 7 exit artifact |

## 5.11 Deployment — local compose for dev, Railway services for prod

### 5.11.1 Service graph

| Service | Local (docker compose) | Railway (one service each) | Notes |
|---|---|---|---|
| postgres | postgres:16-alpine + volume | Railway Postgres plugin | Alembic migrates on api start (kept from v2) |
| redis | redis:7-alpine | Railway Redis plugin | Queue + pub/sub + cache |
| api | uvicorn, port 8000 | Dockerfile.api | Health at /api/v1/health; CORS allowlist = frontend origin |
| worker | rq worker | Dockerfile.worker (same image as api) | Frozen model triplet baked into the image at /app/models; MODEL_VERSION env selects it (F-26; Railway filesystems are ephemeral, so no mounted ./models) |
| frontend | vite dev server | Static build (Railway static or nginx image) | Built with the public API URL (F-27) |

### 5.11.2 Environment variables (the F-27 fix lives here)

| Variable | Local dev | Production | Note |
|---|---|---|---|
| DATABASE_URL / REDIS_URL | compose service names | Railway plugin references | Server-side only — containers can resolve service names |
| VITE_API_BASE | /api (vite proxy to localhost:8000) | https://<api>.up.railway.app/api | Baked into the JS bundle at build time; must be browser-reachable — never a Docker hostname (F-27) |
| VITE_WS_BASE | ws://localhost:8000/ws | wss://<api>.up.railway.app/ws | wss in production or browsers block mixed content |
| MODEL_VERSION | dev-latest | e.g. v3.0-stageC-seed1 | Part of every cache key (E-15) and every API response |
| WANDB_API_KEY | set locally for training only | ABSENT | Training never runs in serving (F-26); v2 had it in the api container |

Reproducibility pack (A-09): make ingest / make fit / make train / make eval / make report; a seeds.yaml manifest; config hashes in every result row; a frozen data snapshot archived once ingestion completes.

## 5.12 Validation protocol — what each metric measures, and how it lies

Every metric ships with its definition, its target, and a written account of how it can be gamed or misread — plus the mitigation. This table is the honest core of Chapter 7 and directly answers your brief.

| Metric | Definition and measurement | Target | How it can be gamed / misread -> mitigation |
|---|---|---|---|
| M1 Paired position delta | Mean finishing-position delta vs the rule-based baseline over 500 paired seeds (identical event draws), bootstrap 95% CI | Mean at least +1.0 with CI excluding 0 | Weak rivals inflate it -> rivals fitted to data, evaluation uses the held-out rival band (E-06); also reported vs replayed-real strategies (M2) |
| M2 Counterfactual vs actual | The 10 v2 races, era-matched (F-24): agent strategy vs the team's actual strategy replayed in the same simulator, same seeds | Mean at least +1.0 (v2's +1.5 kept as stretch) | Sim optimism: the agent optimised the very simulator it is scored in, real teams optimised reality -> framed as "in-simulator counterfactual under stated assumptions", read alongside M8 sim-to-real error |
| M3 Oracle regret | Gap in finishing time/position to a full-information dynamic-programming strategist on the deterministic twin of each eval race | Report distribution; no hard target in run 1 | An easy oracle flatters the agent -> oracle definition published (search space: all pit-lap x compound plans x greedy energy); regret reported with the oracle's assumptions |
| M4 Energy utilisation | Deployed / E_dep on green laps; derate events per race | At least 0.90; derates under 5% of laps | Trivially maxed by always attacking -> only meaningful jointly with M1/M2; reported as a pair with derate rate |
| M5 SC/VSC decision quality | Against the ex-post afterstate optimum per neutralisation window: precision AND recall of "pit now" calls, VSC and SC separated (E-03) | Both at least 0.75 | Always-pit inflates recall -> precision reported jointly; window definition (denominator) published — this replaces v2's single gameable "78% exploitation" number, and the +2.5 reward that pre-paid it is gone (F-17) |
| M6 Pit window accuracy | Share of pits within 3 laps of the post-hoc optimal lap (afterstate search), kept from v2 with the optimum now defined | At least 0.72 (v2 kept) | Optimum depends on the search's information set -> information set stated (no future event knowledge) |
| M7 Strategy diversity | Entropy of (stop count x first-stop compound) across 1000 eval races, per circuit | Report (v2's 2.1 nats dropped as arbitrary) | Dithering fakes diversity -> sanity metric only, never a headline |
| M8 Sim-to-real | Section 5.4.1 suite on held-out 2026 rounds | KS under 0.15 per circuit; pit loss within 1.0 s | Calibrating on the test rounds -> split frozen now (E-14) |
| M9 Tyre failure rate | Thermal failures per 100 eval races | Under 2.0 (fixes v2's reversed "> 2.0") | Trivially zero by never stretching stints -> read jointly with M1 and stint-length distributions |
| M10 Regulatory legality | 100k-random-action property test + zero illegal actions in all eval logs | 100% by construction (A-01) | Nothing to game — that is the point of moving compliance out of the reward (F-16) |
| M11 Luck share (A-08) | Variance decomposition: fix the policy, vary event seeds vs vary decision noise; share of finishing-position variance attributable to event draws | Report — a novel analytics finding | Depends on event-model realism -> event rates validated per circuit first (5.4.1) |

## 5.13 Recruiter demo — the first 90 seconds, engineered

A hiring manager decides in the first minute whether this is a template project or an engineer's project. The opening sequence is therefore specified as tightly as the API.

| Time | What they see | What it proves |
|---|---|---|
| 0-15 s | URL opens directly into a Live Simulation already mid-race (deep link with seed + circuit in the query string): 22 cars moving, agent car highlighted, SOC bar draining and refilling | Deployed, fast, real system — not a video; energy dynamics visible immediately |
| 15-45 s | Lap ~18: the decision card slides in — PIT probability, top-5 SHAP drivers, the counterfactual flip line, and the fidelity badge (e.g. "surrogate agreement 93%") | Explainability with stated honesty — the badge is what separates engineering from marketing |
| 45-75 s | One click to the Counterfactual view: split screen, agent vs the real 2024 strategy, divergence flash at the pit lap, position-delta chart ticking upward | Quantified value against reality, era-matched, with the assumptions printed on screen |
| 75-90 s | One click to the Energy view: SOC trajectory against the per-circuit harvest cap; hover a lap where the model banked energy before an attack | Correct 2026 physics — the exact place a motorsport engineer probes first, and where v2 would have failed |

The v2 ten-minute script (Section 14.2) is kept as the long-form tour with four edits: the ERS Landscape beats become the Energy view beats; "Verstappen's car" becomes the three-letter data code; the W&B beat adds the masking-vs-penalty ablation; the closing GitHub beat shows the reproducibility Makefile.

# Part 6 — Beginner build guide (plain language, step by step)

You said to guide you like you are new to all of this — so here is the whole build as nine phases. Each phase says what it is for, what "done" looks like, and the kind of instruction to give Claude Code. Rule of thumb: **one phase at a time, tests before features, commit when green.** The dates assume you start Wednesday 8 July 2026 and want everything frozen by mid-November (confirm the real deadline in OQ-3; v2's plan started in June, so the calendar below is the honest re-baseline).

## 6.1 How to drive Claude Code with this document

- Put this PDF's Part 5 into the repo as docs/spec/ (one markdown file per section is ideal), and create a CLAUDE.md at the repo root that says: which section is being built now, where the configs live (configs/*.yaml), and the house rules (tests first; no unexplained constants; every physics number carries a source comment).
- Work in packages: one package = one Part 5 subsection + its tests. Example instruction: "Read docs/spec/5.5-environment.md. Write the unit tests for 5.5.5 termination and the 5.5.2 mask rules first, then implement F1RaceEnv until they pass."
- Ask for the test to fail first, then the implementation. If Claude Code invents a constant, make it move the constant to the yaml with a source comment — that habit is what keeps the interpretability contract true.
- Keep sessions scoped: a fresh session per package with CLAUDE.md + the one spec section beats one giant session with everything.

## 6.2 The nine phases

| Phase (dates) | What it is for | "Done" means | Example instruction to Claude Code |
|---|---|---|---|
| 0 — Skeleton (8-11 Jul) | A repo that runs before it does anything: compose (postgres, redis, api hello-world, vite hello-world), CI, lockfiles, folder structure from v2 Section 3.2 + worker/ + configs/ | docker compose up is green; CI passes on push; lockfiles committed | "Set up the repo per docs/spec/5.9 and 5.11: compose file, Dockerfiles, GitHub Actions running pytest and vitest, pinned lockfiles." |
| 1 — Data in (8-21 Jul) | Ingest 2022-2026 FastF1 + OpenF1 into Postgres with the cleaning contract; produce the per-season data-quality reports | Every expected event loaded; quality reports written; completeness assertions pass (E-12) | "Implement docs/spec/5.3: resumable ingest with FastF1 cache and OpenF1 throttling; emit data_quality_report.md per season." |
| 2 — Physics fits + circuits (20 Jul - 4 Aug) | Fit tyre/pace/fuel/event models per circuit (joint fit, E-10); build all circuit ribbons; run T1-T8 and calibrate on 2026 rounds 1-6 | physics_params.yaml + circuits.yaml populated with sources; ribbon geometry tests pass; 5.4.1 validation plots exist for rounds 7+ | "Implement docs/spec/5.4 transformation T4 with the calibration notebook; every coefficient goes to physics_params.yaml with a source comment." |
| 3 — Environment (5-25 Aug) | F1RaceEnv per 5.5 with masks, dynamics, reward, era config; scripted baselines and the oracle strategist; measure throughput | Property tests green (100k random actions: legality, energy conservation, SOC bounds); 22 cars finish; steps/s measured and recorded (5.6.2) | "Write the 5.5.2 mask tests and the E-01 energy property test first, then implement the env until green." |
| 4 — Training (26 Aug - 15 Sep) | Stages A to C with competence gates; the 4 ablations; 3-seed headline run; model selection on validation seeds | Gates logged; best checkpoint chosen; M1/M5/M9 computed on test seeds. **This is the MDP gate — see 6.3** | "Implement docs/spec/5.6-5.7: MaskablePPO training script with W&B logging, gate evaluation on frozen seeds, checkpoint + VecNormalize saving (F-23)." |
| 5 — Explainability (16-22 Sep) | Distill the surrogate, report fidelity, generate decision cards + flip probes for all eval races | Fidelity at least 90% or surrogate deepened; cards stored in agent_decisions; gradient cross-check computed | "Implement docs/spec/5.8; fail the build if surrogate top-1 agreement is below 0.90." |

| Phase (dates) | What it is for | "Done" means | Example instruction to Claude Code |
|---|---|---|---|
| 6 — Backend + first deploy (23 Sep - 6 Oct) | API + worker + WebSocket per 5.9; deploy the skeleton to Railway early so deployment risk dies now, not in November | All endpoints green in integration tests; a full sim round-trip streams over WS; public health URL live | "Implement docs/spec/5.9: RQ worker running the env with the frozen model, playback-clock streaming per E-16, cache keys per E-15." |
| 7 — Frontend (7 Oct - 3 Nov) | Views in value order: 1+2 (weeks 1-2), 3+4 (week 3), 5+6 (week 4); perf budget checks as you go | Each view meets its 5.10 row; Chrome trace of a 70-lap replay holds 45 fps minimum | "Build View 2 per docs/spec/5.10: one InstancedMesh for 22 cars, max 4 Html labels, generic liveries." |
| 8 — Evaluation + freeze (4-17 Nov) | Full 5.12 metric suite; 10 era-matched counterfactuals; sim-to-real chapter numbers; demo video; README; final deploy | Every M1-M11 number exists with CIs; make report regenerates the headline table; demo deep-link works cold | "Run docs/spec/5.12 end to end and generate the Chapter 7 tables + figures via make report." |

*Write as you go: draft Chapter 6 (data/synthetic) during Phase 2, Chapter 3-4 (method/design) during Phases 3-4, Chapter 7 shell during Phase 5. Writing everything in November is how dissertations die.*

## 6.3 The Minimum Defensible Product and the cut order

**MDP (end of Phase 4):** corrected-physics environment + masked PPO agent + M1/M5/M8/M9/M10 with confidence intervals + decision cards rendered in a notebook. That alone is a passing MSc dissertation with genuine novelty (corrected 2026 energy model, masking architecture, sim-to-real validation). Everything after is distinction polish and recruiter material.

If you fall behind, cut in this order — and never cut below the line:

- 1. Views 3 and 4 become 2D Recharts panels (the data pipeline behind them is unchanged).
- 2. View 5 live streaming becomes playback of a precomputed run (the WS plumbing already supports this via E-16).
- 3. Ablations 3 and 4 drop to 1 seed each.
- 4. View 6 split viewport becomes a single canvas with an overlay position-delta chart.
- **Never cut:** action masking, the cleaning contract, the calibration/validation split, the metric suite with its gaming notes, decision cards. These are the dissertation.

# Part 7 — Open questions (your answers unblock the build)

Answer these in one message; none needs more than a line. Defaults apply if you simply say "use defaults".

| ID | Question | Why it matters | Default if unanswered |
|---|---|---|---|
| OQ-1 | Final project name: F1SIP or SENNA? | Repo, domain, dissertation title | F1SIP (matches the submitted spec) |
| OQ-2 | What compute do you actually have? (laptop CPU cores / GPU, university HPC access, Colab?) | Sets training + ablation ceilings (5.6.2) | 12-core CPU assumption; 8M-step budget |
| OQ-3 | Exact COMP702 submission date, and the written AI-use policy | The whole calendar in 6.2, and F-30 compliance | Freeze by 17 Nov; AI use declared |
| OQ-4 | Has your supervisor seen and approved the v3 corrections (physics fixes, scope cuts)? | The proposal said "core idea fixed" — these changes are within that, but get the nod in writing | Proceed, flag at next supervision |
| OQ-5 | Will you pay for an OpenF1 real-time account? | Only affects optional live-weekend overlays | No — live view stays simulator-based (R-05) |
| OQ-6 | Railway plan and monthly budget? | Worker memory/CPU limits; whether frontend is Railway-static or nginx | Hobby plan; static frontend |
| OQ-7 | Keep v2's Stage-B circuit six (Monza, Barcelona, Silverstone, Spa, Suzuka, Monaco)? | Curriculum design (5.7) | Yes, keep |
| OQ-8 | Confirm one decision per lap (vs three per lap by sector) | Sector-level triples episode length and compute for marginal strategic value | Per lap |
| OQ-9 | Counterfactuals: quick fine-tune on 2024-era constants (recommended, ~2-3 days) or 2026-only with a stated caveat? | F-24; affects Chapter 7 framing | Era-matched fine-tune |
| OQ-10 | W&B account/entity name to log under? | Phase 4 logging | Personal free tier |
| OQ-11 | How much of v2's June plan already exists (repo? ingested data? anything)? | Re-baselining: Phase 0-1 may compress | Assume nothing exists |
| OQ-12 | Keep v2's target-team table for demo tailoring? | Phase 8 demo emphasis per team | Yes, keep |

# Part 8 — Assumptions register (2026 physics and data)

Every load-bearing assumption, graded: A = verified against two or more primary sources this week; B = single good source; C = must be fitted from data or verified before the phase that uses it. If a row changes, the config file changes — never the code.

| ID | Assumption | Grade | Source | If wrong -> action |
|---|---|---|---|---|
| AR-01 | MGU-K max 350 kW (v2's 400 kW was the ICE) | A | Formula1.com 2026 PU explainers; FIA 2026 PU regs | Update constants yaml; refit T1 |
| AR-02 | Deployment taper: full to 290 km/h, zero at 355 km/h | A | Formula1.com explainer | Refit E_dep integrals |
| AR-03 | Manual Override: 350 kW to 337 km/h, ~0.5 MJ, within 1.0 s | A | Formula1.com; ESPN 2026 guide | Adjust override feature + T7 |
| AR-04 | Energy Store usable capacity 4 MJ (unchanged) | A | ESPN 2026 guide; cross-check FIA Art. 5.4 | Change SOC bound everywhere via config |
| AR-05 | Harvest cap 8.5 MJ/lap baseline; FIA-adjusted ~5-9 per event; extra quali reductions | A | FIA regs; Raceteq (May 2026); Autosport (Mar 2026) | Per-event H_max values in circuits.yaml |

| ID | Assumption | Grade | Source | If wrong -> action |
|---|---|---|---|---|
| AR-06 | Lift-and-coast harvests at 350 kW and forces Z-mode (drag cost) | B | Raceteq 2026 energy explainer | Adjust conserve-mode cost c_lc |
| AR-07 | Super clipping recovers at ~250 kW on throttle | B | Autosport Mar 2026 | Adjust H_clip term |
| AR-08 | Aero vs 2024: downforce about -30%, drag about -55% | B | Honda 2026 regulations overview | Refit T2 |
| AR-09 | Minimum mass 768 kg (-30 kg) | A | Honda overview; F1 aero explainer | Refit T3 |
| AR-10 | Race fuel mass about 70 kg, energy-flow limited | C — VERIFY before Phase 2 | To confirm vs FIA 2026 regs | Set from the verified figure in T6 |
| AR-11 | Fastest-lap championship point abolished from 2025 | B — 30-second check | 2025/2026 Sporting Regulations | If somehow restored: add back as an eval metric, not a reward |
| AR-12 | 22-car grid (11 teams, Cadillac entered) | A | 2026 season fact | Grid size is a config value |
| AR-13 | 2026 calendar currently 22 rounds; Madrid new in September; roughly 10 rounds raced by 7 July | A | Sky Sports / season reporting | Circuit library is data-driven (F-08) |
| AR-14 | Ergast dead since early 2025; FastF1 >= 3.7 serves the same interface from Jolpica | A | FastF1 3.7 documentation; project announcements | None — v3 already assumes it |
| AR-15 | OpenF1: 2023 onward, free historical at ~3 req/s, real-time paid | A | openf1.org | Throttle values in ingest config |
| AR-16 | FastF1 car/pos telemetry cadence roughly 4-5 Hz (not 10 Hz) | B | FastF1 documentation | Energy integrals are cadence-aware already |
| AR-17 | 2026 dirty-air penalty smaller than 2024 (closer-racing intent) | C — fit | Design assumption | Fit traffic term from 2026 rounds in T7 calibration |
| AR-18 | 2026 Pirelli compound behaviour differs from C1-C5 | C — fit | Pirelli 2026 announcements | T5 refits from rounds 1-6 by design |

### Primary sources consulted for this revision (7 July 2026)

- Formula1.com — "2026 regulations explained: power units" (Jan 2026) and the 2026 PU / aerodynamics explainers (2024); F1 2026 terminology update (Dec 2025).
- FIA — 2026 Formula 1 Power Unit Technical Regulations (Art. 5: ERS-K power, per-lap harvest limits, ES).
- Raceteq — "MGU-K, megajoules and managing the battery: F1's 2026 energy system explained" (May 2026).
- Autosport — "FIA cuts recoverable energy in F1 qualifying" (Mar 2026).
- ESPN — "2026 F1 rules: what's new on the cars" (Mar 2026).
- Honda Global — 2026 Formula 1 Regulations Overview (Jan 2026).
- FastF1 3.7 documentation — Jolpica-F1 API interface; Fast-F1 GitHub discussion #445 (Ergast deprecation).
- openf1.org — API documentation and access-tier notes.
- Sky Sports F1 / season reporting — 2026 calendar status, Cadillac entry, 22-car grid.

## Approval checklist — what happens next

- 1. Read Part 0, then Parts 1, 3 and 4 (the audit and the change lists). Part 5 can wait until the build.
- 2. Answer the twelve questions in Part 7 (one message; "use defaults" is a valid answer for most).
- 3. Show your supervisor Part 1 findings F-01 to F-09 and the Part 3 cut list — five minutes, and it protects you.
- 4. Reply "approved" (with any edits). We then start Phase 0 in Claude Code the same day, using Part 6.1 as the working method.

*End of specification. Version 3.0 — prepared 7 July 2026 — supersedes v2.0 in full. Where v2 was right, it is carried forward by name; where it was wrong, the correction and its source are recorded above.*